CHAPTER 82

Knee

The knee is the largest synovial joint in the body. It consists of three functional compartments that collectively form a dynamic, specialized hinge joint. During propulsion, the knee is able to withstand impressive weight-bearing loads while conducting precision movements, providing a stable yet fluid mechanism for relatively efficient bipedal locomotion. The complex arrangement of intra- and extracapsular ligaments that helps to counter the considerable biomechanical demands that are imposed on the joint can also be involved in disease (i.e. tri-compartment disease).

SKIN AND SOFT TISSUES

SKIN

Innervation

Infrapatellar branch of the saphenous nerve

The infrapatellar branch of the saphenous nerve reaches the anterior aspect of the knee from the medial side. It is invariably divided in the medial surgical approaches to the knee, which accounts for the numbness that inevitably occurs following such procedures. A painful neuroma may form if the nerve is partially sectioned, e.g. by the incision for an arthroscopy portal or a small medial arthrotomy. Unfortunately, the position of the nerve relative to the line of the joint is variable. In most cases, it crosses just below the joint line, passing over the patellar ligament at its insertion on to the tibia. For further details, see Tennant et al (1998).

Peripatellar plexus

Proximal to the knee, the infrapatellar branch of the saphenous nerve connects with branches of the medial and intermediate femoral cutaneous nerves, and lateral femoral cutaneous nerve. Distal to the knee, it connects with other branches of the saphenous nerve. This fine, subcutaneous network of communicating nerve fibres over and around the patella is termed the peripatellar plexus.

Cutaneous vascular supply and lymphatic drainage

The arterial supply of the skin covering the knee is derived from genicular branches of the popliteal artery, the descending genicular branch of the femoral artery, and the anterior recurrent branch of the anterior tibial artery, with small contributions from muscular branches to vastus medialis and the posterior thigh muscles (**Fig. 82.1**). For further details, consult Cormack and Lamberty (1994).

Cutaneous veins are tributaries of vessels that correspond to the named arteries. Cutaneous lymphatic drainage is initially to the superficial inguinal nodes, possibly also to the popliteal nodes, and then to the deep inguinal nodes.

SOFT TISSUES

Popliteal fossa

The popliteal fossa (**Figs 82.2**–**82.3**) is a diamond-shaped region posterior to the knee, bordered by posterior compartment muscles of the thigh and leg. The boundaries are biceps femoris proximolaterally; semimembranosus and the overlying semitendinosus proximomedially; the lateral head of gastrocnemius with the underlying plantaris distolaterally; and the medial head of gastrocnemius distomedially. The anterior boundary (or floor) of the fossa is formed, in proximodistal sequence, by the popliteal surface of the femur, the oblique popliteal ligament (overlying the posterior surface of the capsule of the knee joint), and the posterior aspect of the proximal tibia covered by popliteus and its fascia. The deep fascia (fascia musculorum) acts as the roof of the fossa and is continuous with the fascia lata proximally and with the deep fascia of the leg distally. It is a dense layer that is strongly reinforced by transverse fibres and is often perforated by the short saphenous vein and medial and lateral sural cutaneous nerves; these structures are useful landmarks in the direct posterior approach to the knee joint.

The popliteal fossa is approximately 2.5 cm wide. Distally, its contents are protected and hidden by the heads of gastrocnemius, which contact each other. The fossa contains the popliteal vessels (see Fig. 82.2; **Fig. 82.4**), tibial and common fibular nerves, short saphenous vein, medial and lateral sural cutaneous nerves, posterior femoral cutaneous nerve, articular branch of the obturator nerve, lymph nodes, fat and a variable number of bursae. The tibial nerve descends centrally immediately anterior to the deep fascia, crossing the vessels posteriorly from lateral to medial. The common fibular nerve descends laterally immediately medial to the tendon of biceps femoris. When the popliteal vessels enter the proximal region of the popliteal fossa, they maintain a side-by-side relationship, which shifts to an over–under relationship as they descend through the fossa and are held together by dense areolar tissue within the fossa. This may potentially compromise the popliteal artery in distal femoral fractures. The popliteal vein is generally located posterior to the artery. Proximally, the vein lies lateral to the artery, crossing to its medial side distally. At times, the popliteal vein may be duplicated, so that the artery lies between the veins, which are usually bridged by connecting channels. An articular branch from the obturator nerve descends along the artery to the knee. Six or seven popliteal nodes are embedded in the fat, one under the deep fascia near the termination of the short saphenous vein, one between the popliteal artery and knee joint, and the others intimate with the popliteal vessels.

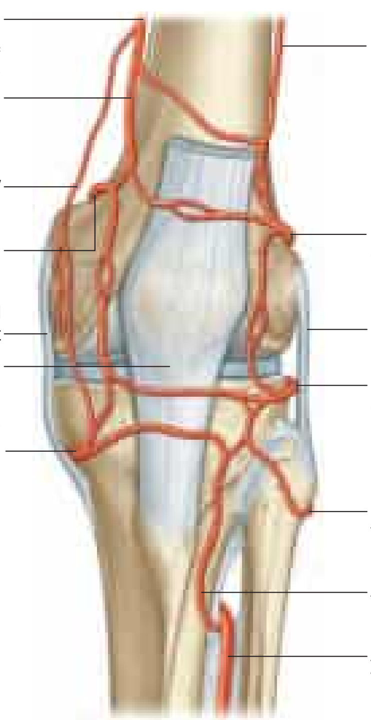


Fig. 82.1 The arterial anastomoses around the left knee joint, anterior aspect.

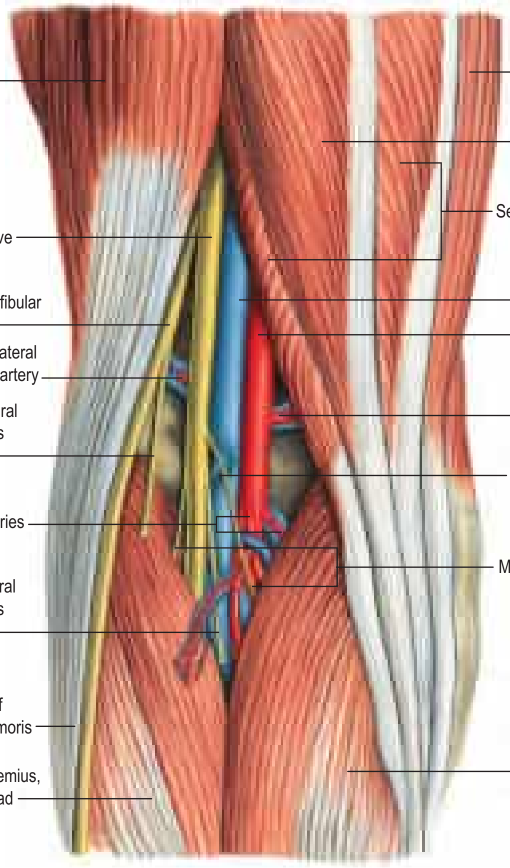


Fig. 82.2 The left popliteal fossa. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

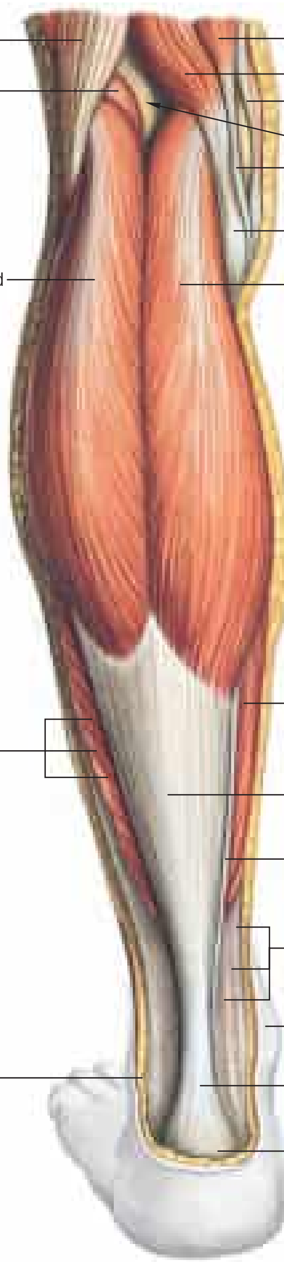


Fig. 82.3 Muscles of the calf, superficial view including the boundaries of the popliteal fossa. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

BONES

FEMUR, TIBIA AND FIBULA

The femur is described on pages 1348–1353 and the tibia and fibula are described on pages 1401–1405 and 1405, respectively.

PATELLA

The patella is the largest sesamoid bone in the body (Figs 82.5–82.6) and is embedded in the tendon of quadriceps femoris, lying anterior to the distal femur (femoral condyles). It is flat, distally tapered and proximally curved, and has anterior and articular surfaces, three borders and an apex, which is the distal end of the bone. Most surfaces and borders are palpable. With the knee in extension, the apex is positioned proximal to the line of the knee joint by 1–2 cm.

The subcutaneous, convex anterior surface is perforated by numerous nutrient vessels. It is longitudinally ridged, separated from the skin by the subcutaneous prepatellar bursa, and covered by an expansion from the tendon of quadriceps femoris, which blends distally with superficial fibres of the patellar ligament (inaccurately named because this structure is the continuation of the tendon of quadriceps femoris). The posterior surface has a proximal smooth, oval articular area, crossed by a smooth vertical ridge, which fits the intercondylar groove on the femoral patellar surface and divides the patellar articular area into medial and lateral facets; the lateral is usually larger. Each facet is divided by faint horizontal lines into approximately equal thirds. A seventh 'odd' facet is present as a narrow strip along the medial border of the patella; it contacts the medial femoral condyle in extreme knee flexion. Distal to the articular surface, the apex is roughened by the attachment of the patellar ligament. Proximal to this, the area between the roughened apex and the articular margin is covered by an infrapatellar fat pad. The patellar articular cartilage is the thickest in the body, reflecting the magnitude of the stresses to which it is subjected.

The thick superior border (surface) slopes anteroinferiorly. The medial and lateral borders are thinner and converge distally; the expansions of the tendons of vasti medialis and lateralis (medial and lateral patellar retinacula, respectively) are attached to them. The lateral retinaculum receives contributions from the iliotibial tract. Ossification occasionally extends from the lateral margin of the patella into the tendon of vastus lateralis.

The shape of the patella can vary and certain configurations are associated with patellar instability. Not infrequently, a bipartite and, less commonly, a tripartite patella are seen on imaging. The bone seems to be in separate parts, usually with a smaller superolateral fragment: this has long been attributed to the presence of a separate ossification centre but, in some cases, could represent failed union following either a stress fracture or a violent contraction of quadriceps femoris (e.g. landing on the feet after jumping from a substantial height) resulting in a traumatic fracture.

Structure The patella consists of more or less uniformly dense trabecular bone, covered by a thin compact lamina. Trabeculae beneath the anterior surface are parallel to the surface; elsewhere, they radiate from the articular surface into the substance of the bone.

Muscle attachments Quadriceps femoris is attached to the superior surface, except near its posterior margin; the attachment extends distally on to the anterior surface. The attachment for rectus femoris is anteroinferior to that for vastus intermedius. Rough markings can be traced in continuity around the periphery of the bone from the anterosuperior surface to the deep surface of the apex. Those at the lateral and medial

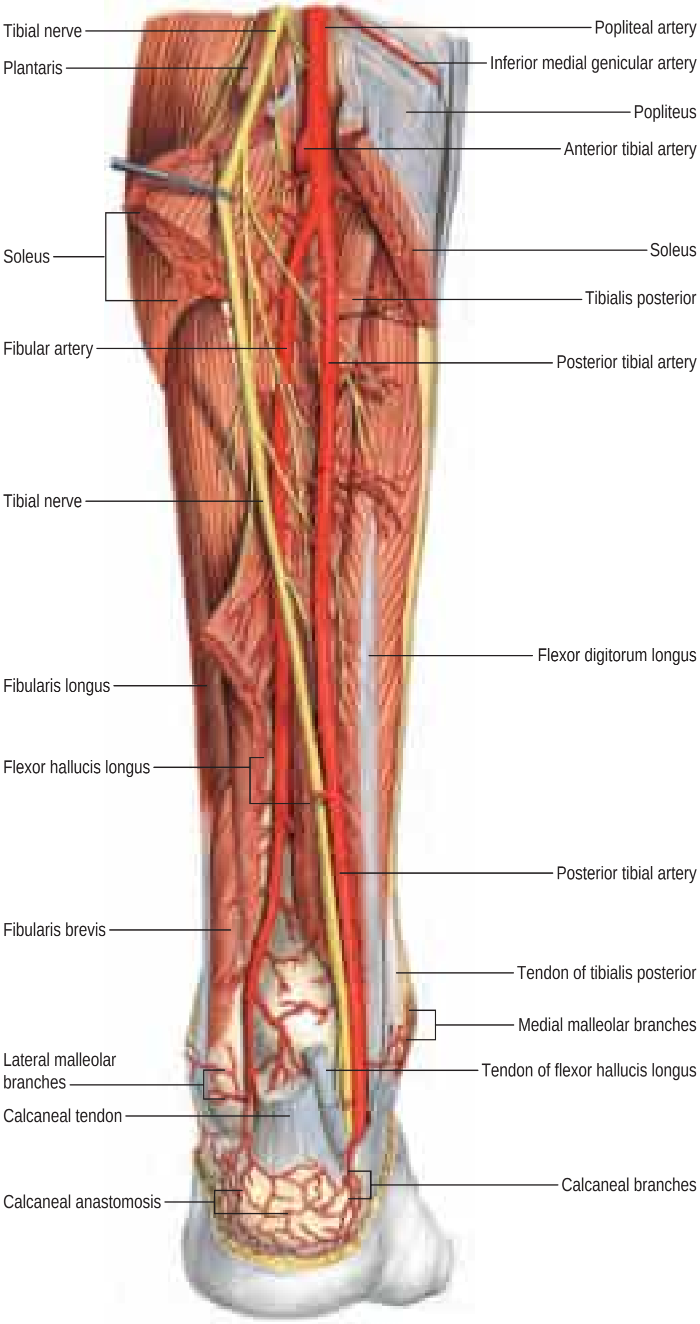


Fig. 82.4 The left popliteal, posterior tibial and fibular arteries, posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

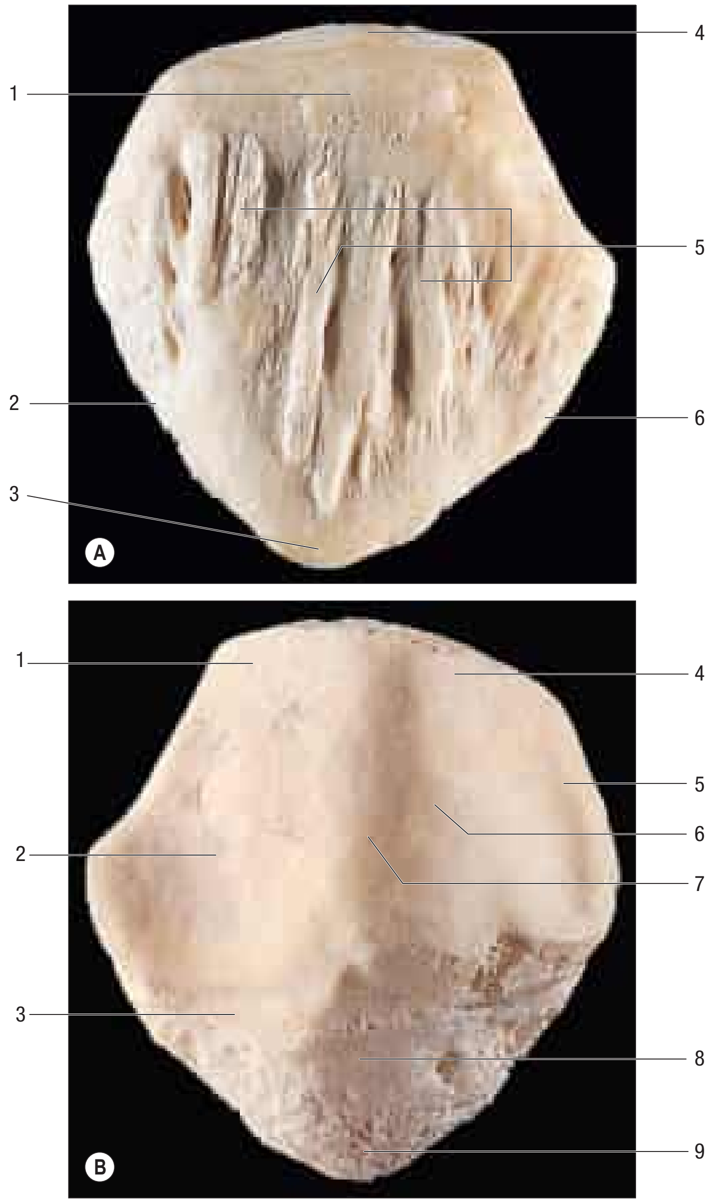


Fig. 82.5 **A**, Left patella, anterior aspect. Key: 1, area of attachment of rectus femoris; 2, medial border: attachment of medial retinaculum (expansion); 3, apex; 4, area of attachment of vastus intermedius; 5, markings of attachment of tendon of quadriceps femoris; 6, lateral border: attachment of lateral retinaculum (expansion). **B**, Left patella, articular (posterior) surface. Key: 1, upper lateral facet: in contact with femur in flexion of the knee; 2, lower lateral facet: in contact with femur in extension of the knee; 3, area overlain by edge of circumferential fat pad; 4, upper medial facet: in contact with femur in flexion of the knee; 5, medial vertical ('odd') facet: in contact with femur in extreme flexion of the knee; 6, lower medial facet: in contact with femur in extension of the knee; 7, ridge; 8, area covered by infrapatellar fat pad; 9, area for attachment of patellar ligament.

borders represent the attachments of vasti lateralis and medialis, and those at the apex represent the attachment of the patellar ligament.

Vascular supply The arterial supply of the patella is derived from the genicular anastomosis, particularly from the genicular branches of the popliteal artery and from the anterior tibial recurrent artery. An anatomical study in children and fetuses confirmed that this network is already well developed in these age groups (Hamel et al 2012).

Ossification Several centres appear during the third to sixth years and these coalesce rapidly. Accessory marginal centres appear later and fuse with the central mass.

JOINTS

SUPERIOR TIBIOFIBULAR JOINT

The superior (proximal) tibiofibular joint is a synovial joint (plane variety) between the lateral tibial condyle and head of the fibula.

Articulating surfaces The articulating surfaces vary in size, form and inclination. The joint line may be transverse or oblique (in the latter case, the joint surfaces are inclined at an angle of greater than 20°). The fibular facet is usually elliptical or circular, and almost flat or slightly grooved. The surfaces are covered with hyaline cartilage.

The volume of articular cartilage peaks at Tanner stage 2; boys gain articular cartilage faster than girls. The rate of cartilaginous volume development is +233 µl/year for the patella, +350 µL/year for the medial tibial compartment and +256 µL/year for the lateral tibial compartment (Jones et al 2003).

Fibrous capsule The capsule is attached to the margins of the articular surfaces of the tibia and fibula, and is thickened anteriorly and posteriorly.

Ligaments The ligaments of the superior tibiofibular joint are not entirely separate from the capsule. The anterior ligament is made up of two or three flat bands, which pass obliquely up from the fibular head to the front of the lateral tibial condyle in close relation to the tendon of biceps femoris. The posterior ligament is a thick band that ascends

Fig. 82.6 Anteroposterior (**A**) and lateral (**B**) radiographs of the knee of a boy aged 14 years. Key: 1, patella; 2, cartilaginous growth plates; 3, intercondylar eminence; 4, prolongation of proximal tibial epiphysis and growth plate forming the tibial tuberosity.

obliquely between the posterior aspect of the fibular head and the lateral tibial condyle, covered by the popliteal tendon.

Synovial membrane The synovial membrane of the superior tibiofibular joint is occasionally continuous with that of the knee joint via the subpopliteal recess.

Vascular supply and lymphatic drainage The superior tibiofibular joint receives an arterial supply from the anterior and posterior tibial recurrent branches of the anterior tibial artery. Lymphatics follow the arteries and drain to the popliteal nodes.

Innervation The superior tibiofibular joint is innervated by branches from the common fibular nerve and from the nerve to popliteus.

Factors maintaining stability Stability of the superior tibiofibular joint is maintained by the fibrous capsule and the anterior and posterior ligaments, assisted by the biceps femoris tendon and the interosseous membrane of the leg.

Movements Very little movement other than limited gliding takes place at the superior tibiofibular joint. Some movement must occur in conjunction with movement at the inferior tibiofibular joint; however, surgical fusion (arthrodesis) of the superior tibiofibular joint seems to have no effect on movements of the ankle joint.

Relations The common fibular nerve runs posterior to the head of the fibula, medial to the tendon of biceps femoris, which is closely associated with the anterior capsule. The anterior and posterior tibial branches of the popliteal artery, and the fibular artery are all vulnerable inferomedial to the joint.

PATELLOFEMORAL JOINT

The patellofemoral joint is a synovial joint and is part of the knee joint.

Articulating surfaces The articular surface of the patella is adapted to that of the femur. The latter extends on to the anterior surfaces of both femoral condyles like an inverted U. Since the whole area is concave transversely and convex in the sagittal plane, it is an asymmetrical sellar surface. The 'odd' facet on the articular surface of the patella contacts the anterolateral aspect of the medial femoral condyle in full flexion, when the highest lateral patellar facet contacts the anterior part of the lateral femoral condyle. As the knee extends, the middle patellar facets contact the lower half of the femoral surface; in full extension, only the lowest patellar facets are in contact with the femur. In summary, on flexion, the patellofemoral contact point moves proximally and the contact area broadens to cope with the increasing stress that accompanies progressive knee flexion.

Patellar ligament sheath and patellar ligament The patellar ligament is a continuation of the tendon of quadriceps femoris and therefore is inaccurately named. It continues from the patella to the tibial tuberosity (see Figs 80.28, 80.31). It is strong, flat and 6–8 cm in length. Proximally, it is attached to the apex of the patella and adjoining margins, to roughened areas on the anterior surface and to a depression on the distal posterior patellar surface. Distally, it is attached to the superior smooth area of the tibial tuberosity. This attachment is oblique, and is more distal laterally. Its superficial fibres are continuous over the patella with the tendon of quadriceps femoris, the medial and lateral parts of which descend, flanking the patella, to the sides of the tibial tuberosity, where they merge with the fibrous capsule as the medial and lateral patellar retinacula. The patellar ligament is separated from the synovial membrane by a large infrapatellar fat pad and from the tibia by a bursa, and lies within its own well-defined sheath.

In the procedure of tibial osteotomy, the tibia is cut transversely just above the insertion of the patellar ligament. Failure to appreciate the obliquity of the tibial attachment of the tendon may lead to inadvertent division of the tendon during this procedure. The middle third of the patellar ligament may be harvested for surgical repair of a cruciate ligament.

All other aspects of the patellofemoral joint are described with the tibiofemoral joint.

TIBIOFEMORAL JOINT

The tibiofemoral joint is a complex synovial joint and is part of the knee joint.

Articulating surfaces

Proximal tibial surface

The proximal tibial surface (unofficially referred to as the tibial plateau) slopes posteriorly and downwards relative to the long axis of the shaft (Fig. 82.7). The tilt, which is maximal at birth, decreases with age, and

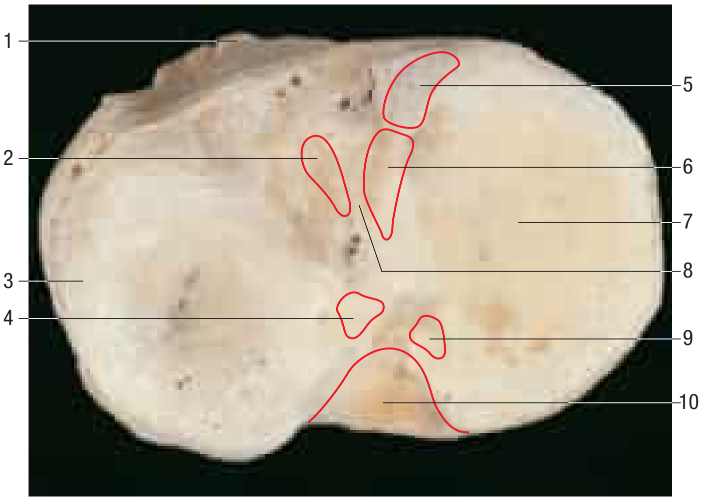


Fig. 82.7 The left tibial plateau. Key: 1, tibial tuberosity; 2, attachment of anterior horn, lateral meniscus; 3, lateral condyle; 4, attachment of posterior horn, lateral meniscus; 5, attachment of anterior horn, medial meniscus; 6, attachment of anterior cruciate ligament; 7, medial condyle; 8, intercondylar eminence; 9, attachment of posterior horn, medial meniscus; 10, attachment of posterior cruciate ligament.

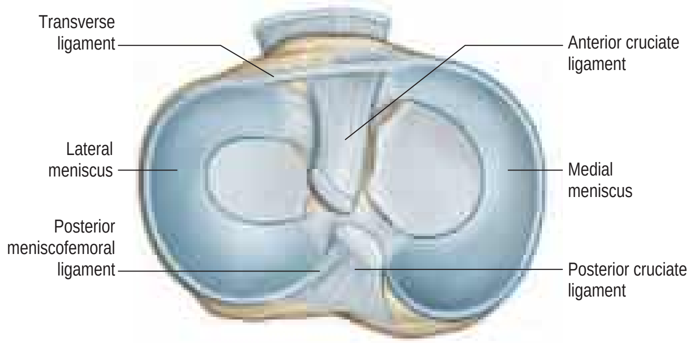


Fig. 82.8 The superior aspect of the left tibia, showing the menisci and the attachments of the cruciate ligaments. (With permission from Drake RL, Vogl AW, Mitchell A, et al (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

Fig. 82.9 Left knee joint, T1-weighted coronal magnetic resonance image (MRI). Key: 1, posterior cruciate ligament; 2, tibial collateral ligament; 3, medial meniscus; 4, lateral meniscus; 5, fibular collateral ligament; 6, anterior cruciate ligament.

is more marked in habitual squatters. The tibial plateau presents medial and lateral articular surfaces (facets) for articulation with the corresponding femoral condyles. The posterior surface, distal to the articular margin, displays a horizontal, rough groove to which the capsule and posterior part of the tibial collateral ligament are attached. The anteromedial surface of the medial tibial condyle is a rough strip, separated from the medial surface of the tibial shaft by an inconspicuous ridge. The medial patellar retinaculum is attached to the medial tibial condyle along its anterior and medial surfaces, which are marked by vascular foramina.

The medial articular surface is oval (long axis anteroposterior) and longer than the lateral articular surface. Around its anterior, medial and posterior margins, it is related to the medial meniscus; the meniscal imprint, wider posteriorly and narrower anteromedially, is often discernible. The surface is flat in its posterior half and the anterior half slopes superiorly about 10°. The meniscus covers much of the posterior surface so that, overall, a concave surface is presented to the medial femoral condyle. Its lateral margin is raised as it reaches the intercondylar region.

The lateral tibial condyle overhangs the shaft of the tibia posterolaterally above a small circular facet for articulation with the fibula. The lateral articular surface is more circular and coapted to its meniscus. In the sagittal plane, the articular surface is fairly flat centrally, and anteriorly and posteriorly the surface slopes inferiorly. Overall, this creates a rather convex surface so that, with the lateral femoral condyle in contact, there are anterior and posterior recesses (triangular in section), which are occupied by the anterior and posterior meniscal 'horns'. Elsewhere, the surface has a raised medial margin that spreads to the lateral intercondylar tubercle. Its articular margins are sharp, except posterolaterally, where the edge is rounded and smooth: here the tendon of popliteus is in contact with bone.

Intercondylar area

The rough-surfaced area between the condylar articular surfaces is narrowest centrally where there is an intercondylar eminence, the edges of which project slightly proximally as the lateral and medial intercondylar tubercles. The intercondylar area widens behind and in front of the eminence as the articular surfaces diverge (see Fig. 82.7; **Fig. 82.8**).

The anterior intercondylar area is widest anteriorly. Anteromedially, anterior to the medial articular surface, a depression marks the site of attachment of the anterior horn of the medial meniscus. Behind this, a smooth area receives the anterior cruciate ligament. The anterior horn of the lateral meniscus is attached anterior to the intercondylar eminence, lateral to the anterior cruciate ligament. The eminence, with medial and lateral tubercles, is the narrow central part of the area. The raised tubercles are thought to provide a slight stabilizing influence on the femur. It is believed that the eminence becomes prominent once walking commences and that the tibial condyles transmit the weight of the body through the tibia.

The posterior horn of the lateral meniscus is attached to the posterior slope of the intercondylar area. The posterior intercondylar area inclines down and backwards behind the posterior horn of the lateral meniscus. A depression behind the base of the medial intercondylar tubercle is for the attachment of the posterior horn of the medial meniscus. The rest of the area is smooth and provides attachment for the posterior cruciate ligament, spreading back to a ridge to which the capsule is attached.

In a study of Nigerian children, the mean intercondylar distance was 0.2 cm at 1 year of age and there was no significant increase in this distance by the age of 10 years (Omololu et al 2003).

Femoral surface

The femoral condyles, bearing articular cartilage, are almost wholly convex. Opinions as to the contours of their sagittal profiles tend to vary. One view is that they are spiral with a curvature increasing posteriorly ('a closing helix'), that of the lateral condyle being greater. An alternative view is that the articular surface for contact with the tibia on the medial femoral condyle describes the arcs of two circles. According to this view, the anterior arc makes contact with the tibia near extension and is part of a virtual circle of larger radius than the more posterior arc, which makes contact during flexion. The lateral femoral condyle is believed to describe a single arc and thus to possess a single radius of curvature.

Tibiofemoral congruence is improved by the menisci, which are shaped to produce concavity of the surfaces presented to the femur; the combined lateral tibiomeniscal surface is deeper. The lateral femoral condyle has a faint groove anteriorly, which rests on the peripheral edge of the lateral meniscus in full extension. A similar groove appears on the medial condyle but does not reach its lateral border, where a narrow strip contacts the medial patellar articular surface in full flexion. These grooves demarcate the femoral patellar and condylar surfaces. The differences between the shapes of the articulating surfaces correlate with the movements of the knee joint.

Menisci

The menisci (semilunar cartilages) are crescentic, intracapsular, fibrocartilaginous laminae (see Fig. 82.8; **Fig. 82.9**). They serve to widen, deepen and prepare the tibial articular surfaces that receive the femoral condyles. Their peripheral attached borders are thick and convex, and their free, inner borders are thin and concave. Their peripheries are vascularized by capillary loops from the fibrous capsule and synovial membrane, while their inner regions are less vascular. Tears of the menisci are common. Peripheral tears (e.g. in the vascularized zone) have the potential to heal satisfactorily, especially with surgical intervention. Tears in the less vascular or inner zones seldom heal spontaneously; if surgery is indicated, these menisci are often resected. The meniscal horns are richly innervated compared with the remainder of the meniscus. The central one-third is devoid of innervation (Gronblad et al 1985). The proximal surfaces are smooth and concave, and in contact with the articular cartilage on the femoral condyles. The distal surfaces are smooth and flat, resting on the tibial articular cartilage. Each covers approximately two-thirds of its tibial articular surface. Canal-like structures open on to the surface of the menisci in infants and young children, and may transport nutrients to deeper, less vascular areas.

Two structurally different regions of the menisci have been identified. The inner two-thirds of each meniscus consist of radially organized collagen bundles, and the peripheral one-third consists of larger circumferentially arranged bundles (Ghadially et al 1983). Thinner collagen bundles parallel to the surface line the articular surfaces of the inner part, while the outer portion is covered by synovium. This structural arrangement suggests specific biomechanical functions for the two regions: the inner portion of the meniscus is suited to resisting compressive forces while the periphery is capable of resisting tensional forces. With ageing and degeneration, compositional changes occur within the menisci, which reduce their ability to resist tensional forces. Outward displacement of the menisci by the femoral condyles is resisted by firm anchorage of the peripheral circumferential fibres to the intercondylar bone at the meniscal horns.

The menisci spread load by increasing the congruity of the articulation, provide stability by their physical presence and proprioceptive feedback, and may cushion the underlying bone from the considerable forces generated during extremes of flexion and extension of the knee.

Medial meniscus

The medial meniscus is broader posteriorly and is almost a semicircle in shape (see Fig. 82.8). It is attached by its anterior horn to the anterior tibial intercondylar area in front of the anterior cruciate ligament; the posterior fibres of the anterior horn are continuous with the transverse ligament of the knee (when present). The anterior horn is in the floor of a depression medial to the upper part of the patellar ligament. The posterior horn is fixed to the posterior tibial intercondylar area, between the attachments of the lateral meniscus and posterior cruciate ligament. Its peripheral border is attached to the fibrous capsule and the deep surface of the tibial collateral ligament. The tibial attachment of the meniscus is known as the 'coronary or meniscotibial ligament'. Collectively, these attachments ensure that the medial meniscus is relatively fixed and moves much less than the lateral meniscus.

Lateral meniscus

The lateral meniscus forms approximately four-fifths of a circle and covers a larger area than the medial meniscus (see Fig. 82.8). Its breadth, except at its short tapered horns, is more or less uniform. It is grooved posterolaterally by the tendon of popliteus, which separates it from the fibular collateral ligament. Its anterior horn is attached in front of the intercondylar eminence, posterolateral to the anterior cruciate ligament, with which it partly blends. Its posterior horn is attached behind this eminence, in front of the posterior horn of the medial meniscus. Its anterior attachment is contorted so that the free margin faces posterosuperiorly, and the anterior horn rests on the anterior slope of the lateral intercondylar tubercle. Near its posterior attachment, it commonly sends a posterior meniscofemoral ligament superomedially behind the posterior cruciate ligament to the medial femoral condyle. An anterior meniscofemoral ligament may also connect the posterior horn to the medial femoral condyle anterior to the posterior cruciate ligament. The meniscofemoral ligaments are often the sole attachments of the posterior horn of the lateral meniscus. More laterally, part of the tendon of popliteus is attached to the lateral meniscus, and so mobility of its posterior horn may be controlled by the meniscofemoral ligaments and by popliteus. A meniscofibular ligament occurs in most knee joints. As with the medial meniscus, there is a tibial attachment via the so-called coronary ligament, but the meniscus has no peripheral bony attachment in the region of popliteus; in the surgical literature, this gap is referred to as the popliteus hiatus.

Discoid lateral meniscus A discoid lateral meniscus occasionally occurs, often bilaterally. The distinguishing features of a discoid lateral meniscus are its shape and posterior ligamentous attachments. The following classification of the abnormality is based on the work of Watanabe et al (1979). In its mildest form, the partial discoid meniscus is simply a wider form of the normal lateral meniscus. The acute, medial free edge is interposed between the femoral and tibial condyles but it does not completely cover the tibial plateau. A complete discoid meniscus appears as a biconcave disc with a rolled medial edge and covers the lateral tibial plateau. The Wrisberg type of meniscus has the same shape as a complete discoid meniscus but its only peripheral posterior attachment is by the meniscofemoral ligaments. In this case, the normal tibial attachment of the posterior horn of the lateral meniscus is lacking but the posterior meniscofemoral ligament persists. As a result, this type of meniscus is attached anteriorly to the tibia and posteriorly to the femur, which renders the posterior horn unstable. Under these circumstances, the meniscus is liable to become caught between the femur and tibia: this accounts for the classic presenting symptom of the 'clunking knee' in some patients. The aetiology of discoid meniscus is not clear. Most are asymptomatic and are often found by chance at arthroscopy. However, they may cause difficulty in gaining access to the lateral compartment at arthroscopy. A discoid medial meniscus is extremely rare.

Transverse ligament of the knee

The transverse ligament of the knee connects the anterior convex margin of the lateral meniscus to the anterior horn of the medial meniscus (see Fig. 82.8). It varies in thickness and is often absent. Its exact role is conjectural, although one study found that the ligament was slightly taut in knee extension (Tubbs et al 2008); presumably, it helps to decrease tension generated in the longitudinal circumferential fibres of the menisci when the knee is subjected to load. A posterior meniscomeniscal ligament is sometimes present.

Meniscofemoral ligaments

The two meniscofemoral ligaments connect the posterior horn of the lateral meniscus to the inner (lateral) aspect of the medial femoral condyle (**Figs 82.10–82.11**). The anterior meniscofemoral ligament (ligament of Humphrey) passes anterior to the posterior cruciate ligament. The posterior meniscofemoral ligament (ligament of Wrisberg) passes behind the posterior cruciate ligament and attaches proximal to the margin of attachment of the posterior cruciate.

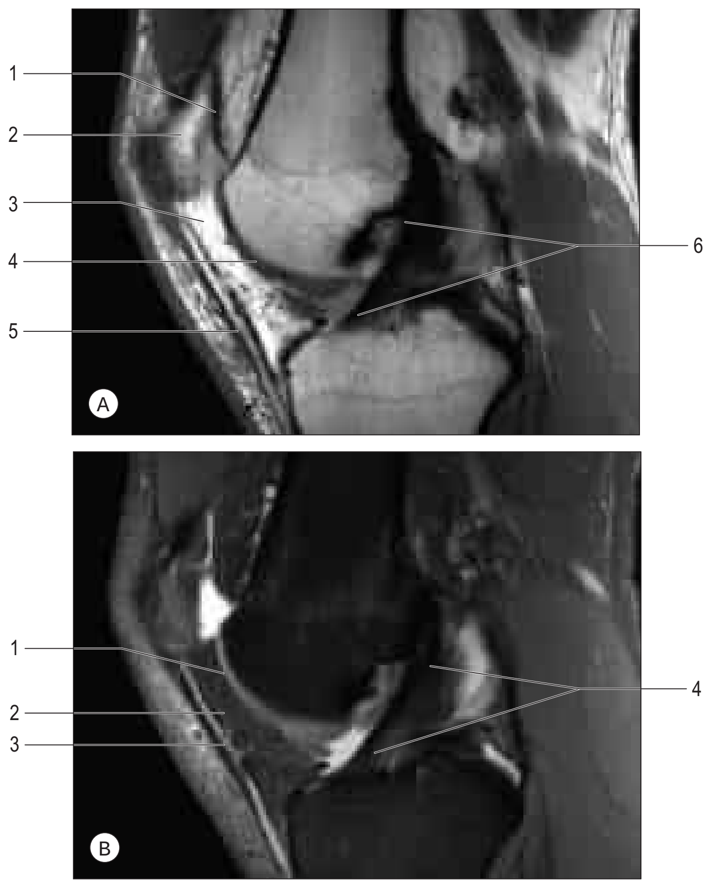


Fig. 82.10 A, The anterior cruciate ligament, proton density (PD)-weighted sagittal MRI. Key: 1, suprapatellar bursa; 2, patella; 3, infrapatellar fat pad; 4, femoral articular cartilage; 5, patellar ligament; 6, anterior cruciate ligament. **B**, Anterior cruciate ligament: T2-weighted sagittal MRI. Key: 1, femoral articular cartilage; 2, infrapatellar fat pad; 3, patellar ligament; 4, anterior cruciate ligament.

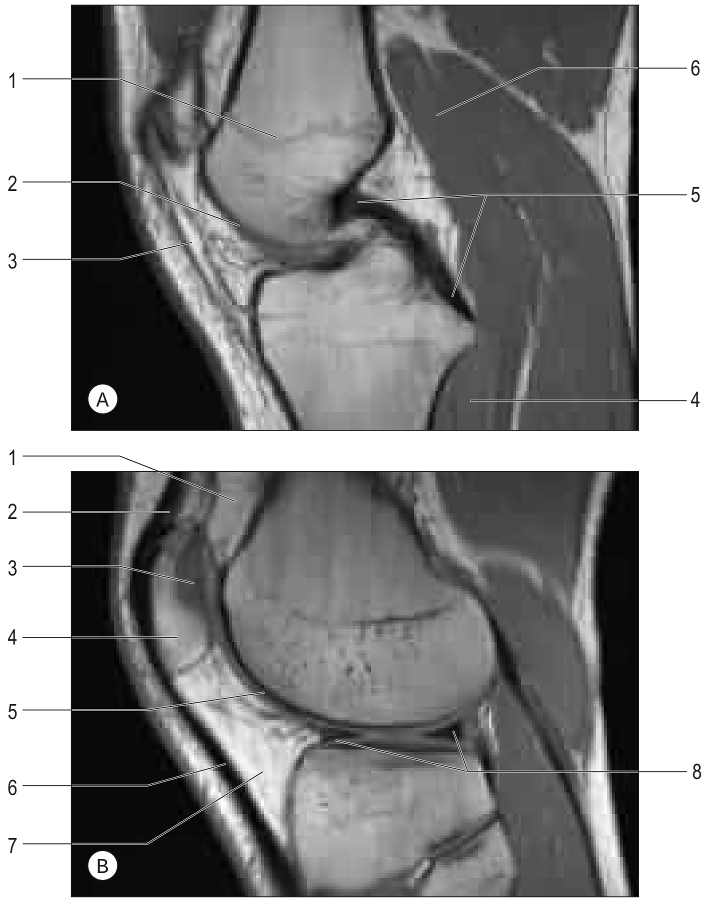


Fig. 82.11 **A**, Posterior cruciate ligament, PD-weighted sagittal MRI. Key: 1, epiphyseal line; 2, femoral articular cartilage; 3, patellar ligament; 4, popliteus; 5, posterior cruciate ligament; 6, gastrocnemius. **B**, Medial meniscus, PD-weighted sagittal MRI. Key: 1, suprapatellar fat pad; 2, tendon of quadriceps femoris; 3, patellar articular cartilage; 4, patella; 5, femoral articular cartilage; 6, patellar ligament; 7, infrapatellar fat pad; 8, medial meniscus.

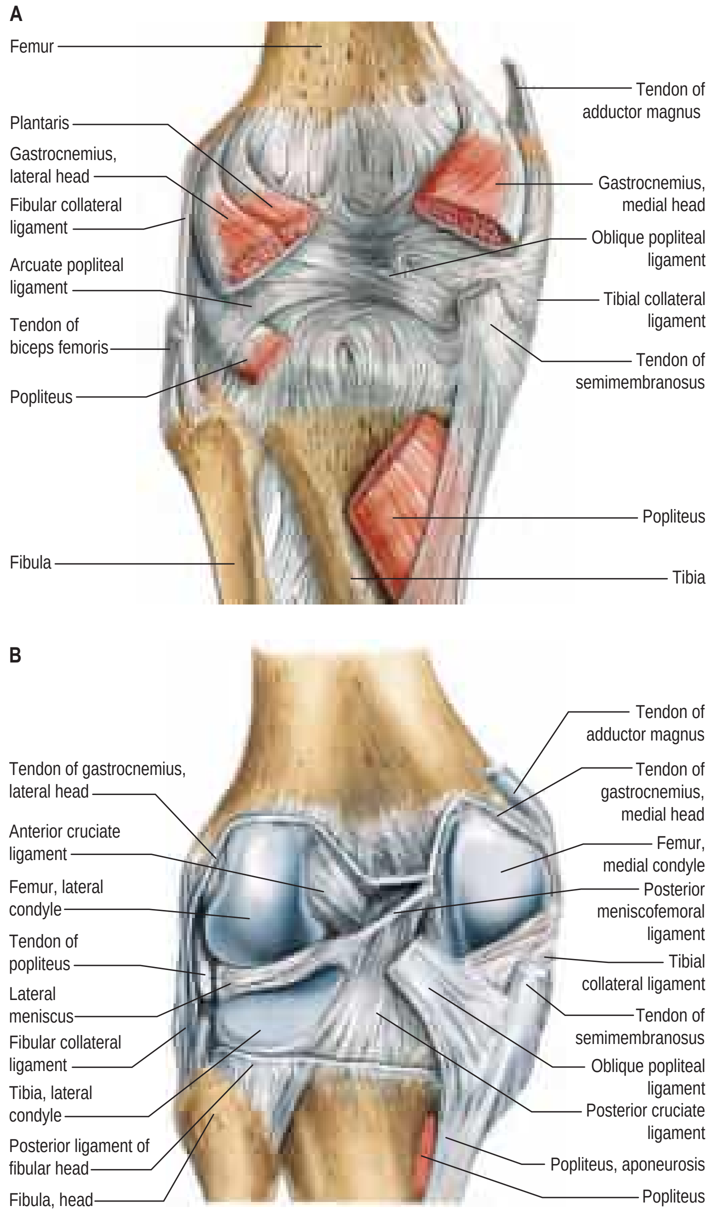


Fig. 82.12 A posterior dissection of the knee. **A**, Capsule intact. **B**, Capsule removed. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Anatomical studies found that at least one meniscofemoral ligament was almost always present in the cadaveric knees examined, while both sometimes coexisted (Gupte et al 2003). Biomechanical studies have revealed the cross-sectional area and strength of the meniscofemoral ligaments to be comparable to those of the posterior fibre bundle of the posterior cruciate ligament.

The meniscofemoral ligaments are believed to act as secondary restraints, supporting the posterior cruciate ligament in minimizing displacement caused by posteriorly directed forces on the tibia. These ligaments are also involved in controlling the motion of the lateral meniscus in conjunction with the tendon of popliteus during knee flexion.

Soft tissues

Recent advances in imaging and surgery of knee ligaments have contributed to an improved understanding of the anatomy of the medial and lateral soft tissues of the knee.

Capsule and retinacula

The joint capsule is a fibrous membrane of variable thickness. Anteriorly, it is replaced by the patellar ligament and does not pass proximal to the patella or over the patellar area. Elsewhere, it lies deep to expansions from vasti medialis and lateralis, separated from them by a plane of vascularized loose connective tissue. The expansions are attached to the patellar margins and patellar ligament, extending back to the corresponding collateral (tibial and fibular) ligaments and distally to the tibial condyles. They form medial and lateral patellar retinacula, the lateral being reinforced by the iliotibial tract.

Posteriorly, the capsule contains vertical fibres that arise from the articular margins of the femoral condyles and intercondylar fossa, and from the proximal tibia. The fibres mainly pass downwards and somewhat medially. The oblique popliteal ligament is a well-defined thickening across the posteromedial aspect of the capsule, and is one of the major extensions from the tendon of semimembranosus.

Medial soft tissues

The medial soft tissues (see Fig. 80.28; **Fig. 82.12**) are arranged in three layers (Warren and Marshall 1979).

Layer 1 Layer 1 is the most superficial and is the deep fascia that invests sartorius. The saphenous nerve and its infrapatellar branch are superficial to the deep fascia of the leg. Sartorius inserts into the fascia as an expansion rather than as a distinct tendon. The fascia spreads inferiorly and anteriorly to lie superficial to the distinct and readily identifiable tendons of gracilis and semitendinosus and their insertions. The latter two tendons are commonly harvested for surgical reconstruction of damaged cruciate ligaments. To gain access to them, the upper edge of sartorius can be identified. The sartorius (layer 1) fascia is then incised to reveal the tendons. Deep to the tendons is the anserine bursa, which overlies the superficial part of the tibial collateral ligament; this bursa sometimes becomes inflamed, especially in track and field athletes. Posteriorly, layer 1 overlies the tendons of gastrocnemius and the structures of the popliteal fossa. Anteriorly, layer 1 blends with the anterior limit of layer 2 and the medial patellar retinaculum. More inferiorly, layer 1 blends with the periosteum.

A condensation of tissue passes from the medial border of the patella to the medial epicondyle of the femur (the medial patellofemoral ligament), the anterior horn of the medial meniscus (the meniscopatellar ligament), and the medial tibial condyle (the patellotibial ligament).

Layer 2 Layer 2 is the plane of the superficial part of the tibial collateral ligament, which means that the tendons of gracilis and semitendinosus lie between layers 1 and 2. The superficial part of the tibial collateral ligament has vertical and oblique portions. The former contains vertically orientated fibres that pass from the medial epicondyle of the femur to a large insertion on the medial surface of the proximal end of the tibial shaft. It extends to an area about 5 cm distal to the joint line. Its anterior edge is rolled and easily seen just posterior to the insertions of gracilis and semitendinosus once layer 1 has been opened. The posteriorly placed oblique fibres run posteroinferiorly from the medial epicondyle of the femur to blend with the underlying layer 3 (capsule), effectively to insert on the posteromedial tibial articular margin and posterior horn of the medial meniscus. This area is reinforced by a part of the insertion of semimembranosus. There is a vertical split in layer 2 anterior to the superficial part of the tibial collateral ligament. The fibres anterior to the split pass superiorly to blend with vastus medialis fascia and layer 1 in the medial patellar retinaculum. The fibres posterior to the split pass superiorly to the medial epicondyle and thence anteriorly as the medial patellofemoral ligament.

Layer 3 Layer 3 is the capsule of the knee joint and can be separated from layer 2 everywhere except anteriorly close to the patella, where it blends with the more superficial layers. Deep to the superficial part of the tibial collateral ligament it is thick and has vertically orientated fibres that make up the deep medial part of the tibial collateral ligament. It sends fibres to the medial meniscus. Anteriorly, the separation of the superficial and deep parts of the tibial collateral ligament is distinct. Posteriorly, layers 2 and 3 blend to form a conjoined posteromedial capsule.

Lateral soft tissues

The lateral soft tissues (see Fig. 82.12; **Fig. 82.13**) are also arranged in three layers (Seebacher et al 1982), which collectively have been referred to as the lateral collateral ligamentous complex (Nissman et al 2008). The most superficial layer is the lateral patellar retinaculum. The middle layer consists of the fibular collateral popliteofibular, fabellofibular and arcuate ligaments. The recently described anterolateral ligament of the knee may exist in this layer (Claes et al 2013). The deep layer is the lateral part of the capsule.

The lateral patellar retinaculum consists of superficial oblique and deep transverse portions. The former runs from the iliotibial tract to the patella. The latter is thicker and subdivided into three parts: the lateral patellofemoral ligament, running from the lateral patellar border to the lateral epicondyle of the femur; the transverse retinaculum, running from the iliotibial tract to the mid-patella; and the patellotibial band, running from the patella to the lateral tibial condyle.

The fascia lata and the iliotibial tract lie posterior to the lateral retinaculum. They come together distally to insert on to the tibia at a tubercle (Gerdy's tubercle) on the anterolateral proximal tibia; some fibres continue to insert on the tibial tuberosity. Proximally, the fascia lata merges with the lateral intermuscular septum. Posteriorly, it blends with the fascia over biceps femoris. Here, as it emerges from behind the biceps femoris tendon, the common fibular nerve lies in a thin layer of fat bound by the fascia.

The fibular collateral ligament arises from the lateral epicondyle of the femur posterior to the popliteus insertion and just proximal to the groove for popliteus. It is a cord-like structure that passes distally, superficial to the popliteus tendon and deep to the lateral retinaculum, to attach to the fibular head, where it blends with the biceps femoris tendon just anterior to the apex of the head of the fibula. It is separated from the capsule by a thin layer of fat and the inferior lateral genicular vessels.

The single most important stabilizer of the posterolateral knee is the popliteofibular (short external lateral) ligament. It passes from the popliteus tendon at a level just below the joint line, posteriorly, laterally and inferiorly, to the apex of the head of the fibula. As a passive 'tether' combined with the popliteus tendon proximal to it, it resists lateral rotation of the tibia. Its connection to the tendon of popliteus also allows 'dynamic' tensioning.

The fabellofibular ligament is a condensation of fibres that runs either from the fabella (a sesamoid bone sometimes found within the tendon of the lateral head of gastrocnemius) or from the lateral head of gastrocnemius (if the fabella is absent), to the apex of the head of the fibula. The arcuate ligament is a condensation of fibres that runs from the apex of the head of the fibula, posteromedially over the emerging tendon of popliteus below the level of the tibial joint surface, to the tibial intercondylar area. The lateral joint capsule is thin and blends posteriorly with the arcuate ligament. Anteriorly, it forms the weak, lax coronary or meniscotibial ligament, which attaches the inferior border of the meniscus to the lateral tibia.

Ligaments

Cruciate ligaments

The cruciate ligaments, so named because they cross each other, are very strong, richly innervated intracapsular structures. The point of crossing is located a little posterior to the articular centre. They are named anterior and posterior with reference to their tibial attachments (**Figs 82.14**–**82.15**; see Fig. 82.17). A synovial membrane almost surrounds the ligaments but is reflected posteriorly from the posterior cruciate ligament to adjoining parts of the capsule; the intercondylar part of the posterior region of the fibrous capsule therefore has no synovial covering.

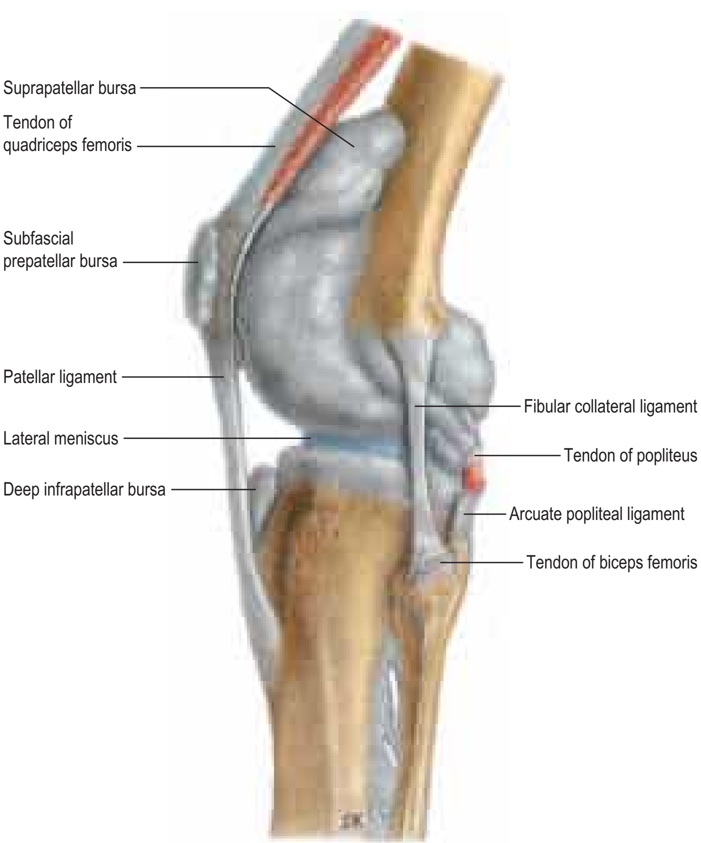


Fig. 82.13 The left knee joint, lateral aspect. The synovial cavity is distended and the synovial membrane appears grey. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

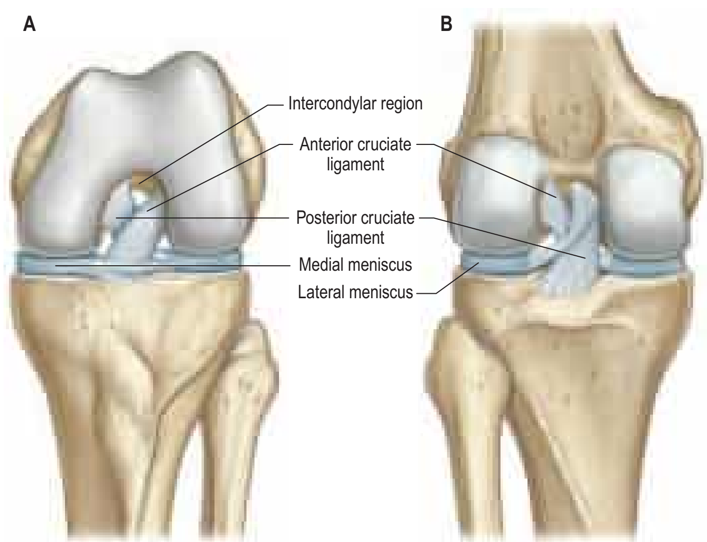


Fig. 82.14 The left knee joint. **A**, Anterior aspect in full flexion. **B**, Posterior aspect in extension. (With permission from Drake RL, Vogl AW, Mitchell A, et al (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

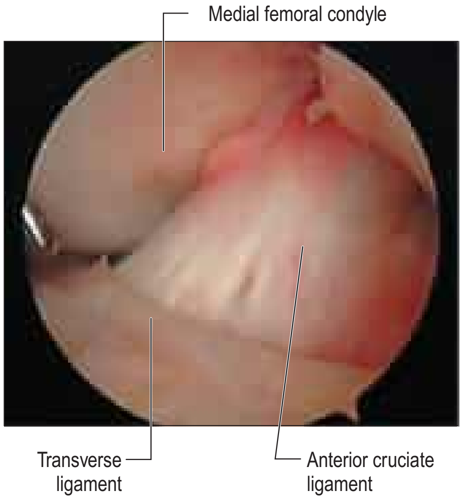


Fig. 82.15 The intercondylar notch at arthroscopy, showing the anterior cruciate and transverse ligaments of the knee. (Courtesy of Smith and Nephew Endoscopy.)

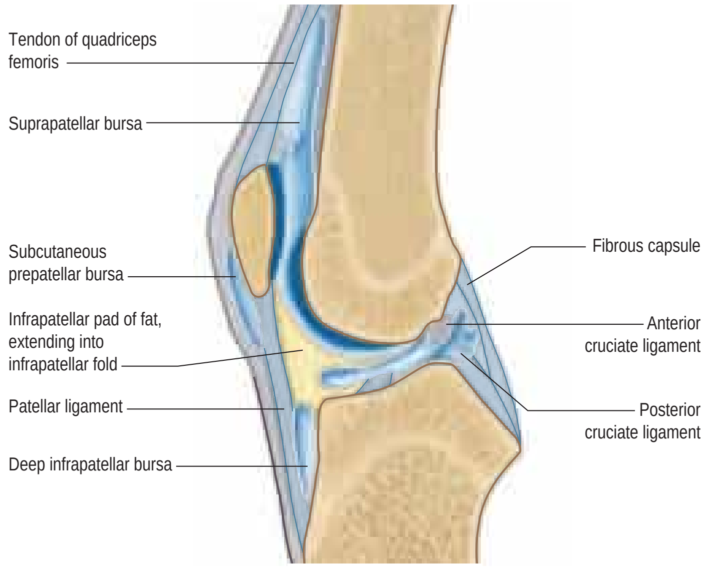


Fig. 82.16 A sagittal section through the left knee joint, lateral aspect.

Anterior cruciate ligament The anterior cruciate ligament is attached to the anterior intercondylar area of the tibia, just anterior and slightly lateral to the medial intercondylar tubercle, partly blending with the anterior horn of the lateral meniscus (see Fig. 82.8). It ascends posterolaterally, twisting on itself and fanning out to attach high on the posteromedial aspect of the lateral femoral condyle (Girgis et al 1975). The average length and width of an adult anterior cruciate ligament are 38 mm and 11 mm, respectively. It is formed of two, or possibly three, functional bundles that are not apparent to the naked eye but can be demonstrated by microdissection techniques. The bundles are named anteromedial, intermediate and posterolateral, according to their tibial attachments (Amis and Dawkins 1991).

Posterior cruciate ligament The posterior cruciate ligament is thicker and stronger than the anterior cruciate ligament (see Fig. 82.8), the average length and width of an adult posterior cruciate ligament being 38 mm and 13 mm, respectively. It is attached to the lateral surface of the medial femoral condyle and extends up on to the anterior part of the roof of the intercondylar fossa, where its attachment is extensive in the anteroposterior direction. Its fibres are adjacent to the articular surface. They pass distally and posteriorly to a fairly compact attachment posteriorly in the intercondylar region and in a depression on the adjacent posterior tibia. This gives a fan-like structure in which fibre orientation is variable. Anterolateral and posteromedial bundles have been defined; they are named (against convention) according to their femoral attachments. The anterolateral bundle tightens in flexion while the posteromedial bundle is tight in extension of the knee. Each bundle slackens as the other tightens. Unlike the anterior cruciate ligament, it is not isometric during knee motion, i.e. the distance between attachments varies with knee position. The posterior cruciate ligament ruptures less commonly than the anterior cruciate ligament and rupture is usually better tolerated by patients than rupture of the anterior cruciate ligament.

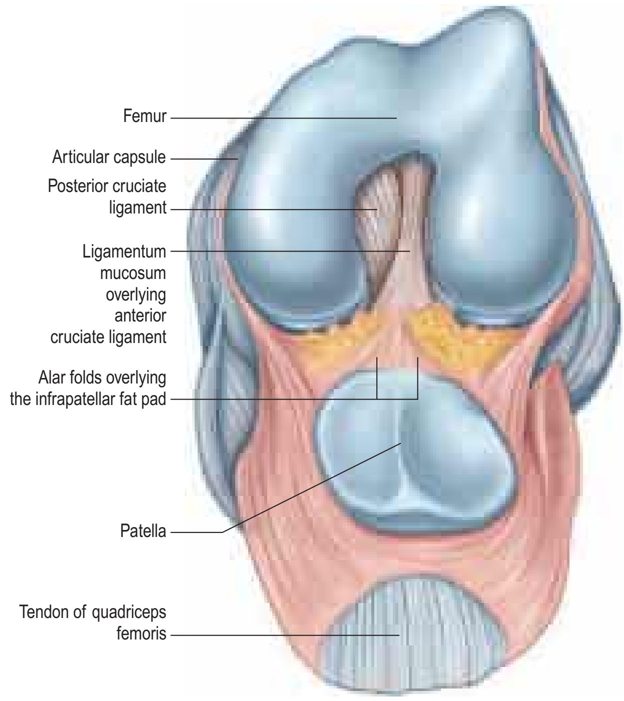


Fig. 82.17 The left knee joint in full flexion. The tendon of quadriceps femoris has been sectioned and the patella retracted distally. Compare with Figure 82.14A.

Synovial membrane, plicae and fat pads

The synovial membrane of the knee is the most extensive and complex in the body. It forms a large suprapatellar bursa between quadriceps femoris and the lower femoral shaft proximal to the superior patellar border (Fig. 82.16). The bursa is an extension of the joint cavity. The attachment of articularis genus to its proximal aspect prevents the bursa from collapsing into the joint. Alongside the patella, the membrane extends beneath the aponeuroses of the vasti, especially under vastus medialis. It extends proximally a hand's breadth above the superior pole of the patella. Distal to the patella, the synovial membrane is separated from the patellar ligament by an infrapatellar fat pad. Where it lies beneath the fat pad, the membrane projects into the joint as two fringes, alar folds, which bear villi. The folds converge posteriorly to form a single infrapatellar fold or plica (ligamentum mucosum), which curves posteriorly to its attachment in the femoral intercondylar fossa (Fig. 82.17). This fold may be a vestige of the inferior boundary of an originally separate femoropatellar joint. The extent of the infrapatellar plica ranges from a thin cord to a complete sheet that can obstruct the passage of instruments during knee arthroscopy. When substantial, it has been mistaken for the anterior cruciate ligament, which is directly posterior to it. The medial plica extends in the midline anteriorly from the medial alar fold medially to the suprapatellar bursa. Occasionally, it can be thickened and inflamed, usually following acute or chronic trauma.

The suprapatellar plicae are remnants of an embryonic septum that completely separates the suprapatellar bursa from the knee joint. Occasionally, a septum persists, either in its entirety or perforated by a small peripheral opening.

The infrapatellar fat pad is the largest part of a circumferential extrasynovial fatty ring that extends around the patellar margins (Newell 1991).

At the sides of the joint, the synovial membrane descends from the femur and lines the capsule as far as the menisci, whose surfaces have no synovial covering. Posterior to the lateral meniscus, the membrane forms a subpopliteal recess between a groove on the meniscal surface and the tendon of popliteus, which may connect with the superior tibiofibular joint. The relationship of the synovial membrane to the cruciate ligaments is described above.

Bursae

Numerous bursae are associated with the knee. Anteriorly, there is a large subcutaneous prepatellar bursa between the lower half of the patella and skin; a small, deep infrapatellar bursa between the tibia and patellar ligament; a subcutaneous infrapatellar bursa between the distal part of the tibial tuberosity and skin; and a large suprapatellar bursa, which is the superior extension of the knee joint cavity (see Fig. 82.16). Posterolaterally, there are bursae between the lateral head of gastrocnemius (lateral subtendinous bursa of gastrocnemius) and the joint capsule (this bursa is sometimes continuous with the joint cavity); the fibular collateral ligament and the tendon of biceps femoris; the fibular collateral ligament and the tendon of popliteus; and the tendon of popliteus and the lateral femoral condyle, which is usually an extension of the synovial cavity of the joint. The last two bursae may communicate with each other.

Medially, the arrangement of the bursae is complex. The bursa between the medial head of gastrocnemius and the fibrous capsule is prolonged between the medial tendon of gastrocnemius and the tendon of semimembranosus (the semimembranosus bursa), and usually communicates with the joint. The bursa between the tendon of semimembranosus and the medial tibial condyle and the medial head of gastrocnemius may communicate with this bursa. The anserine bursa is located between the tibial collateral ligament and the tendons of sartorius, gracilis and semitendinosus. Bursae that vary in both number and position lie deep to the tibial collateral ligament between the joint capsule, femur, medial meniscus, tibia or tendon of semimembranosus. Occasionally, there may be a bursa between the tendons of semimembranosus and semitendinosus. Posteriorly, bursae associated with the knee are variable.

The clinically important bursae are the anterior group, the anserine bursa and the semimembranosus bursa. Inflammation of the subcutaneous prepatellar bursa and infrapatellar bursa are referred to colloquially as 'housemaid's knee' and 'clergyman's knee', respectively. The anserine bursa can become inflamed, especially in athletes. In adults, bursal inflammation producing a popliteal fossa swelling commonly occurs secondary to degeneration within the knee joint; regardless of its size and position, it almost always arises from the plane between semimembranosus and the tendon of the medial head of gastrocnemius.

Relations and 'at risk' structures

The tendon of quadriceps femoris (which encloses and is attached to the non-articular surfaces of the patella), the patellar ligament, tendinous expansions from vasti medialis and lateralis (which extend over the anteromedial and anterolateral aspects of the capsule, respectively), and the patellar retinacula all lie anterior to the knee joint. Posteromedially are sartorius and the tendon of gracilis (which lies along its posterior border); both descend across the joint. Posterolaterally, the tendon of biceps femoris and the common fibular nerve (which lies medial to the tendon) are in contact with the capsule, and thereby separated from the tendon of popliteus. Posteriorly, the popliteal artery and associated lymph nodes lie posterior to the oblique popliteal ligament; the popliteal vein is posteromedial or medial, and the tibial nerve is posterior to both. The nerve and vessels are overlapped by both heads of gastrocnemius and laterally by plantaris. Gastrocnemius contacts the capsules on either side of the vessels. Semimembranosus lies between the capsule and semitendinosus, medial to the medial head of gastrocnemius.

Movements at the knee

Movements at the knee are customarily described as flexion, extension, medial (internal) and lateral (external) rotation. Flexion and extension differ from true hingeing, in that the articular surface profiles of the femoral and tibial articular surfaces produce a variably placed axis of rotation during the flexion arc, and when the foot is fixed, flexion entails corresponding conjunct (coupled) lateral rotation. These conjunct rotations are a product of the complex geometry of the articular surfaces and, to an extent, the disposition of the associated ligaments. There is differential motion in the medial and lateral tibiofemoral compartments. Laterally, there is considerable displacement of the femur on the tibia, with rolling as well as sliding at the joint surface. In contrast, medially, for most of the flexion arc there is minimal relative motion of the femur and tibia, and the motion almost exclusively involves one joint surface sliding on the other. In full flexion, the lateral femoral condyle is close to posterior subluxation off the lateral tibial articular surface. Medially, significant posterior femoral displacement only occurs when flexion exceeds 120°. The menisci move with the femoral condyles, the anterior horns more than the posterior, and the lateral meniscus considerably more than the medial.

The axial rotations have a smaller range than the arc of flexion and extension. These rotations are conjunct, and integral with flexion and extension, i.e. they are obligatory. They can also be adjunct and independent, i.e. voluntary, and are best demonstrated with the knee semiflexed. The degree of axial rotation therefore varies with flexion and extension.

The range of extension is 5–10° beyond the 'straight position'. Active flexion is approximately 120° with the hip extended, 140° when it is flexed, and 160° when aided by a passive element, e.g. sitting on the heels. Voluntary rotation is 60–70° but conjunct rotation only 20°.

Conjunct medial rotation of the femur on the tibia in the later stages of extension is part of a 'locking' mechanism, the so-called 'screw-home movement', which is an asset when the fully extended knees are subjected to strain. Full extension results in the close-packed position, with maximal spiralization and tightening of the ligaments. The roles of the articular surfaces, musculature and ligaments in generating conjunct rotations remain controversial (Girgis et al 1975, Rajendran 1985) but the following points can be made. The lateral combined meniscotibial 'receiving surface' is smaller, more circular and more deeply concave. Since the articular surface is virtually convex in sagittal section, the depth of the receiving surface is largely due to the presence of the lateral meniscus. The lateral femoral articular surface is also smaller. Consequently, the lateral femoral condyle approaches full congruence with the opposed surface some 30° before full extension (well before the medial condyle). Simple extension cannot continue, but medial rotation of the femur occurs on a vertical axis through its head and medial condyle; the medial femoral condyle moves very little in the sagittal plane and is stabilized by the 'upslope' of the anterior half of the medial tibia, while rotation of the lateral femoral condyle and meniscus brings the anterior horn of the latter on to the anterior 'downslope' of the lateral tibial condyle. Rotation and extension follow simultaneously and smoothly until final close packing of both condyles is accomplished. At the beginning of flexion from full extension (with the foot fixed), lateral femoral rotation occurs, which 'unlocks' the joint. While joint surfaces and many ligaments are involved, electromyographic evidence reveals that contraction of popliteus is important, and that it pulls down and backwards on the lateral femoral condyle, lateral to the axis of femoral rotation. It also retracts the posterior horn during lateral rotation and continuing flexion, via its attachment to the lateral meniscus, and so prevents traumatic compression.

Any position of extension adopted is a balance between forces (torque) extending the joint and passive mechanisms resisting them. The range near to close packing is functionally important. In symmetrical standing, the line of the body's weight is anterior to the transverse axes of the knee joints, but the passive mechanisms noted above preserve posture with minimal muscular effort (Joseph 1960). Active contraction of quadriceps femoris and a close-packed position only occurs in asymmetrical postures, e.g. in leaning forward, during heavy loading, or when powerful thrust is needed.

In knee extension, parts of the cruciate ligaments, the tibial and fibular collateral ligaments, the posterior capsular region, the oblique popliteal ligament, skin and fasciae are all taut. Passive and sometimes active tension exists in the posterior thigh muscles and gastrocnemius, and the anterior part of the medial meniscus is compressed between the femoral and tibial condyles. During extension, the patellar ligament is tightened by quadriceps femoris but is relaxed in the erect attitude. When the knee flexes, the fibular collateral ligament and the posterior part of the tibial collateral ligament relax but the cruciate ligaments and the anterior part of the tibial collateral ligament remain taut; the posterior parts of the menisci are compressed between the femoral and tibial condyles. Flexion is checked by quadriceps femoris, anterior parts of the knee joint capsule, posterior cruciate ligament and compression of soft tissues behind the knee. In extreme passive flexion, contact of the calf with the thigh may be the limiting factor and parts of both cruciate ligaments are also taut. In addition to conjunct rotation with terminal extension or initial flexion, relaxed collateral ligaments also allow independent medial and lateral rotation (adjunct rotation) when the joint is flexed.

Accessory movements

Wider rotation can be obtained by passive movements when the knee is semi-flexed. To a limited extent, the tibia can also be translated backwards and forwards on the femur. Abduction and adduction are prevented in full extension by the collateral ligaments and secondary restraints such as the cruciate ligaments. With the knee slightly flexed, limited adduction and abduction are possible, both passive and active. Slight separation of the femur and tibia can be achieved by strong traction on the leg with countertraction applied to the thigh.

Physiological knee joint laxity may occur during puberty. Increased knee joint flexibility is seen more frequently in adolescent girls than boys. There is an inverse relationship between Tanner stage and the degree of laxity with a progressive decrease of sagittal laxity at the onset of Tanner stage 2 (Falciglia et al 2009).

Muscles producing movements

Flexion Flexion is produced by biceps femoris, semitendinosus and semimembranosus, assisted by gracilis, sartorius and popliteus. With the foot stationary, gastrocnemius and plantaris also assist (see Fig. 82.3).

Extension Extension is produced by quadriceps femoris, assisted by tensor fasciae latae.

Medial rotation of the flexed leg Medial rotation of the flexed leg is produced by popliteus, semimembranosus and semitendinosus, assisted by sartorius and gracilis.

Lateral rotation of the flexed leg Lateral rotation of the flexed leg is produced by biceps femoris.

Patellofemoral joint The alignment of the femoral and tibial shafts is such that the pull of quadriceps femoris on the patella imparts a force on the patella that is directed both superiorly and laterally. The static bony factors that counter this tendency to move laterally are the congruity of the patellofemoral joint and the buttressing effect of the larger lateral part of the patellar surface of the femur, which, clinically, is often referred to as the trochlear groove. Instability of the patella may result if the patella is small or if the patellar surface of the femur is too shallow. The static ligamentous factors are the medial patellofemoral ligament and medial patellar retinaculum.

Dynamic neuromuscular control is important. The most distal part of vastus medialis (vastus medialis obliquus) consists of transverse fibres that are attached directly to the medial edge of the patella: these pull the patella medially, countering the tendency to move laterally. This is the muscle that is preferentially strengthened in a physical therapy programme aimed at treating patellofemoral conditions often associated with patella tracking problems such as those seen in growing adolescents.

Tibiofemoral joint The tibiofemoral joint surfaces are inherently mobile, especially laterally. Medially, some stability is afforded by the relatively concave tibial surface and the relatively fixed posterior horn of the medial meniscus. Both medially and laterally the menisci are helpful, particularly as they move with the femoral condyles. Ligaments play a major role in constraining mobility because they bind the bones in positions of extreme stress and also provide proprioceptive feedback, aiding coordination of stabilizing muscle activity. To take a somewhat 'two-dimensional' view, the tibial and fibular collateral ligaments may be considered as sensors (resistors) of valgus and varus forces on the knee, respectively, and the anterior and posterior cruciate ligaments as sensors (resistors) of anterior and posterior tibial translation, respectively. However, in reality the situation is more complex than this. The stresses are rarely applied in orthogonal planes and so a combination of forces, especially rotational, is involved. Moreover, many structures other than the collateral and cruciate ligaments are involved in stabilizing the joint. The 'posterolateral corner', which resists tibial lateral rotation, consists of the popliteofibular, fabellofibular, arcuate and fibular collateral ligaments and iliotibial tract, together with popliteus, the lateral head of gastrocnemius and biceps femoris. The 'posteromedial corner', which resists tibial rotation, consists of the posterior oblique portion of the superficial part of the tibial collateral ligament, the capsule including the oblique popliteal ligament and semimembranosus. Since stresses are often a combination of force plus rotation, structures usually operate together rather than in isolation.

Loading at the knee

During level walking, the force across the tibiofemoral joint for most of the cycle is between two and four times body weight, and can be more. In contrast, the force across the patellofemoral joint is no more than 50% of body weight. Peak force transmission across the joint increases sequentially as the menisci, articular cartilage and subchondral bone are damaged or removed. Walking up or down stairs has little influence on tibiofemoral forces, but significantly increases patellofemoral forces to two (walking up) or three (walking down) times body weight, reflecting the changed angle of the tendon of quadriceps femoris and patellar ligament during flexion. There are two mechanisms for ameliorating forces transmitted across the patella: the extensor lever arm is lengthened as the axis of rotation moves posteriorly during flexion, and the contact area between the patella and femur almost triples between 30° and 90°. To cope with the potential large forces generated by activities such as running, the patella has the thickest articular cartilage in the body.

BIOMECHANICS OF THE KNEE

The knee joint is a complex synovial joint consisting of the tibiofemoral and patellofemoral articulations. It functions to control the centre of body mass and posture in the activities of daily living. This necessitates a large range of movement in three dimensions coupled with the ability to withstand high forces. These conflicting parameters of mobility and stability are only achieved by the interactions between the articular surfaces, the passive stabilizers and the muscles that cross the joint.

The relatively incongruent nature of the joint surfaces makes the knee joint inherently mobile. In addition, because it acts as a pivot between the longest bones in the body, and is subjected to considerable loads in locomotion, the joint is also potentially at risk of injury if any of the multiple factors providing joint stability are compromised. The long bones may act as levers, increasing the stresses on the stabilizing ligaments.

KNEE JOINT KINEMATICS

The surfaces of the tibia and femur are not as conforming as those of the relatively congruent hip joint. Although this variation in geometry permits motion to occur in six degrees of freedom (Fig. 82.18), the primary motion of the knee occurs in the sagittal plane, and a relatively minor degree of movement occurs in the transverse plane. The knee joint may therefore be described simplistically as a modified hinge joint allowing flexion–extension and a measure of rotatory motion. Knee motion is normally defined as starting from 0° (the neutral position), when the tibia and femur are in line in the sagittal plane. Biomechanically, it is important that the knee reaches the neutral position in extension because that allows the leg to support the body weight like a simple strut when the subject is standing still. When the subject is standing upright, if the knee is flexed, the vertical line of action of the body weight passes posterior to the centre of rotation of the knee, tending to cause the body to tilt posteriorly. To counterbalance this, continuous quadriceps femoris contraction is required, causing expenditure of energy.

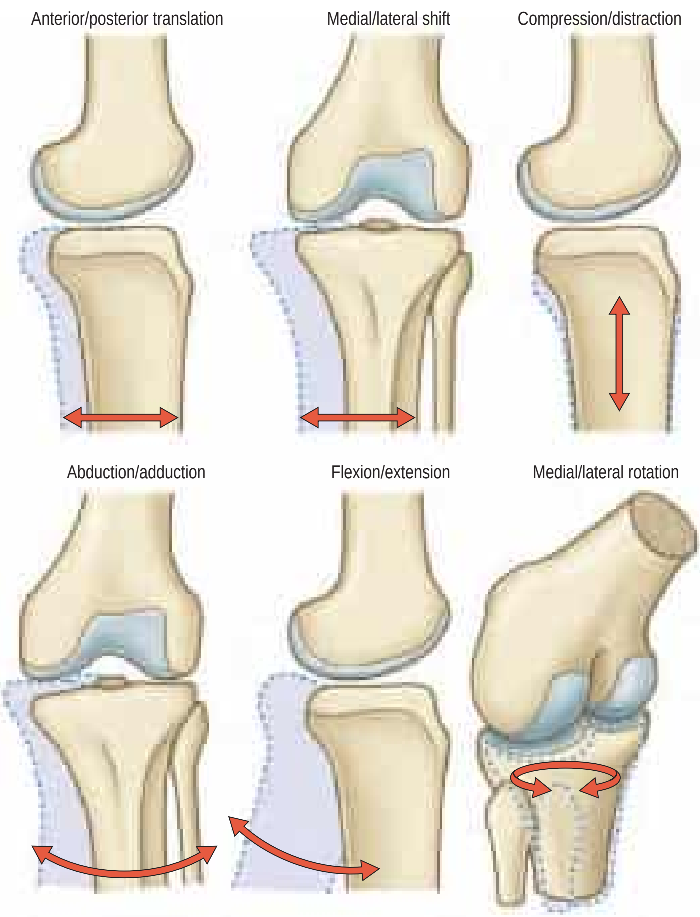


Fig. 82.18 The knee joint motion in three dimensions, described using six independent variables (degrees of freedom).

CHAPTER 82

Active knee flexion leads to approximately 130° flexion. The active motion is limited by apposition of the soft tissue masses (posterior thigh and calf). Passive flexion may reach 160°. This is required in people who habitually kneel as part of daily life, and is a challenge for designers of knee prostheses. It can be observed that such deep flexion is often accompanied by tibial medial rotation, so that, when the subject is kneeling, the buttocks can rest on the feet. This movement carries the lateral tibial plateau anteriorly, so that it does not engage with the femoral lateral condyle in deep knee flexion. The femoral condyle passes posteriorly and rides over the horn of the lateral meniscus.

The most frequently employed knee movement occurs when walking (**Fig. 82.19**). When the leg is swinging past the supporting leg, the knee must be flexed in order to avoid dragging the toes on the ground; this requires approximately 67° knee flexion. When the swinging leg approaches the first contact with the ground, the knee extends, to move the foot forwards for heel strike. If the knee remained extended, this would then cause the body to move in a circular arc, centred at the ankle, causing the centre of gravity to move upwards and then back down again, leading to more energy expenditure. It would also increase the forces on the knee because the leg would act more like a rigid strut, unable to dissipate the impact forces when the foot hit the ground. All these problems are overcome by flexing the knee 15° in the mid-stance phase; the centre of gravity of the body can move forwards at approximately constant height, and the impact energy is absorbed by stretching quadriceps femoris (see Fig. 82.19).

Tibial medial–lateral rotation also occurs during gait: the tibia rotates laterally during terminal extension, a phenomenon known as 'screw-home' (**Fig. 82.20**). It is surmised that this rotation helps to lock the geometry and tighten the soft tissues, thereby maintaining the knee in a stable position prior to the impact load of weight-bearing. The knee acts as one link in a chain of limb segments, and this screw-home relates to rotation of both the foot and hip. When the foot is swung forwards for heel strike, the pelvis rotates so that the hip is moved forwards, a movement that entails lateral rotation of the hip. During stance, the femur is medially rotated against the locked knee. Tibial lateral rotation also causes inversion of the foot at the subtalar joint, raising the arch and locking the structure of the foot. The knee flexion that occurs after the impact on the ground allows the tibia to rotate medially so that the foot everts, softening its structure and allowing it to deform and absorb energy. Conversely, towards toe-off, the knee extends, rotating the tibia laterally, so the foot is again a rigid lever with which to push the body forwards.

Articular kinematics

We now consider knee motion at a smaller scale, within the joint. This, of course, is difficult to separate from the actions of the ligaments that act as passive restraints to tibiofemoral joint movements, and are discussed below.

Sagittal sections of the knee reveal that the arcs of the femoral condyles are much longer than the anterior–posterior length of the tibial plateau. This means that if the knee flexed with a purely rolling motion, then the femur would roll off the back of the tibia long before the knee reaches full flexion (see Fig. 82.20). This does not happen because the femur slides anteriorly at the same time as it rolls posteriorly, and thus remains in correct articulation. If the knee were to possess a fully conforming roller-in-trough geometry, as in the humero-ulnar joint, then flexion would occur by pure sliding movement between the joint

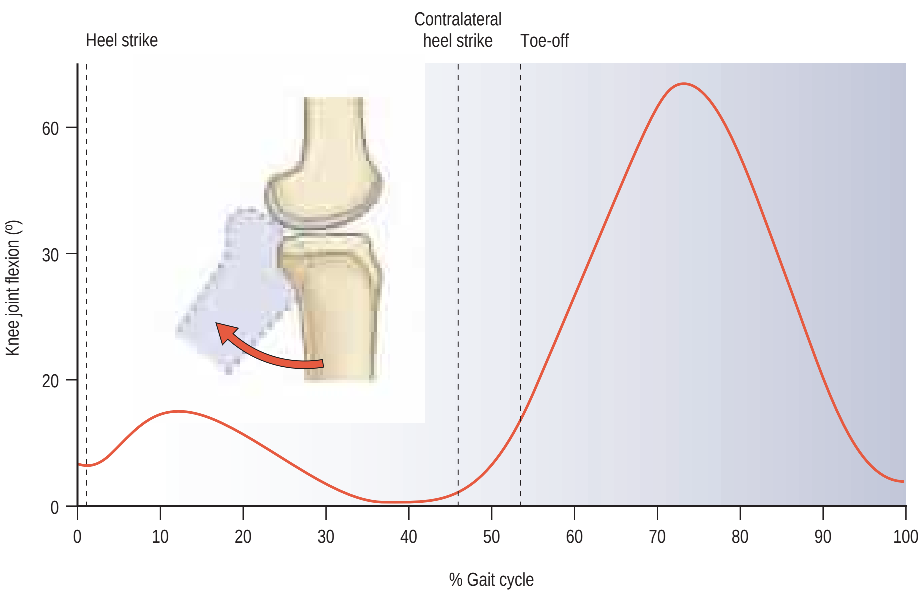


Fig. 82.19 Knee flexion–extension motion during gait.

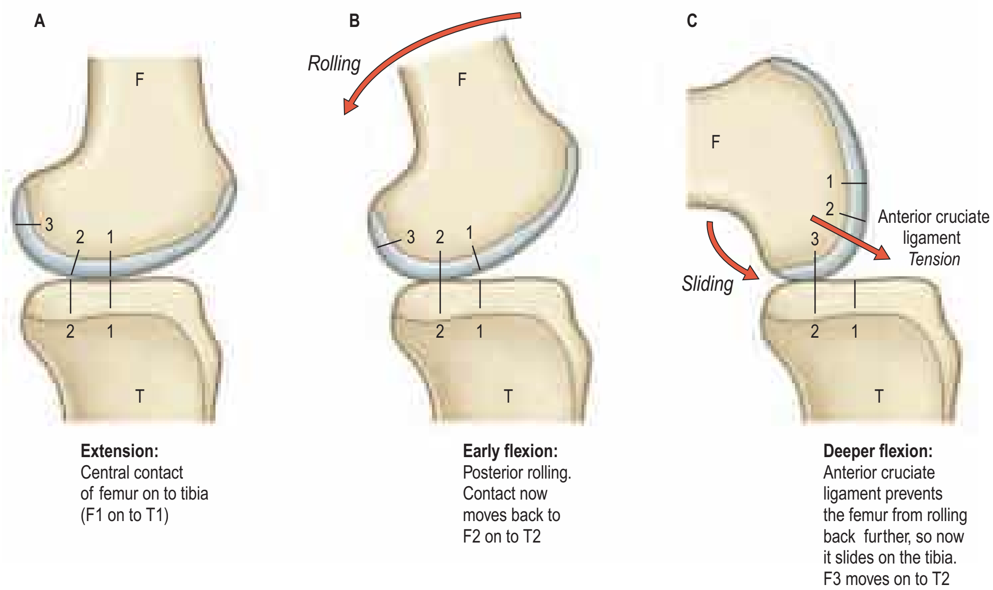


Fig. 82.20 Knee joint kinematics during gait: rolling and sliding (flexion, anterior translation).

surfaces. This conformity is not possible at the knee because it would inhibit the tibial medial–lateral rotation that is needed during locomotion.

In the locked, fully extended knee, the anterodistal femoral articular surfaces press on to the anterior horns of the menisci. This tends to cause the femur to slide posteriorly, tensing the anterior cruciate ligament and slackening the posterior cruciate ligament. With knee flexion, the femur lifts off the anterior horns of the menisci, leading to contact between the smaller radii of the posterior parts of the femoral condyles and the tibial plateau plus the posterior horns of the menisci. This means that the centre of contact moves posteriorly in early knee flexion. After this, the femoral condyles have approximately circular sagittal sections. The femur is now prevented from rolling back any further by tension in the anterior cruciate ligament.

Articular mechanics

The combined effect of the external and internal loads on the knee is to impose considerable forces on the articular cartilage. This may be analysed as a vertical (compressive) component and a horizontal (shear or friction) component. The friction component is a combination of the compressive force and the friction characteristics of the joint surfaces.

Compression

The compressive force is distributed over an area to produce a contact pressure (contact stress). The contact pressure is, therefore, dependent on the area of contact as well as the load itself. Fully conforming articular surface geometry would allow the greatest area of contact, and thus would minimize contact pressure; however, this is not present in the knee. The medial compartment of the knee is semi-conforming with a convex femoral condyle articulating over a concave medial tibial plateau. The lateral compartment has less conformity; the lateral tibial plateau is flat or slightly convex in sagittal sections. These different shapes reflect the differential movement of the medial and lateral compartments. In normal movement, the screw-home rotation of the tibia occurs about a medial axis, which means that most of the rotation of the tibia is due to an anterior–posterior translation of the lateral compartment (Fig. 82.21).

In order to maintain some degree of conformity, and also to minimize contact pressure between the femur and tibia, the menisci are wedge-shaped in cross-section, thereby increasing the area over which the compressive force on the knee is distributed. In the absence of the menisci, the load is carried by a much smaller area of cartilage, resulting in higher contact stresses on the articular cartilage. This helps to explain the prevalence of osteoarthrosis following meniscectomy (Fig. 82.22).

The menisci are most firmly attached at the intercondylar eminence of the tibia, which means that they can translate anteriorly and posteriorly to 'follow' the femoral rollback. The lateral meniscus is more mobile because it is attached to the capsule less tightly than the medial meniscus. The anterior–posterior movement of the menisci is approximately half the magnitude of the anterior–posterior movement of the femur, suggesting that the conformity of the joint changes during flexion of the knee joint.

The distal aspect of the femur, resting on the menisci in the extended knee, has a large radius of curvature and thus fits against the entire area of the menisci. However, as the knee flexes, the smaller radii of the posterior parts of the femoral condyles cause the femur to lift off the anterior horns of the menisci, and the femoral contact is now solely on the posterior horns in the flexed knee. This, combined with the large joint forces that are generated when arising from a seated position, explains why there are often spontaneous tears of the posterior horns in older patients, when standing up from a squatting position.

The load-carrying mode of the menisci is shown in Figure 82.23. When the femoral condyle presses down on the meniscus, it tends to squeeze the meniscus out of the joint because of its tapered cross-section. This causes the diameter of the circular shape of the meniscus (and therefore the meniscal circumference) to increase. This is resisted by 'hoop tension' in strong fibres that pass around the periphery of the meniscus, and transmit the tension to the tibial plateau via strong insertional ligaments. As the tissue is adapted to resist hoop stresses, it has much greater hoop strength (approximately 100 MPa) than radial strength (approximately 3 MPa), which explains why bucket handle tears occur. It also explains why a circumferential tear does not have such a serious effect on meniscal function because the circumferential fibres can still transmit the loads, whereas a radial tear breaks the load-carrying fibres.

Friction

The knee acts as a bearing that transmits forces and movements between the limb segments. Load-bearing synovial joints move with remarkably little friction and their surfaces must withstand many millions of impact loads, which tend to cause fatigue failure and breakdown of the surfaces.

During walking, every step involves phases of action that vary the loading and velocity conditions at the knee joint. A variety of lubrication mechanisms normally prevent joint surface damage. Thus, in the swing phase, when the foot is off the ground, the joint surfaces are

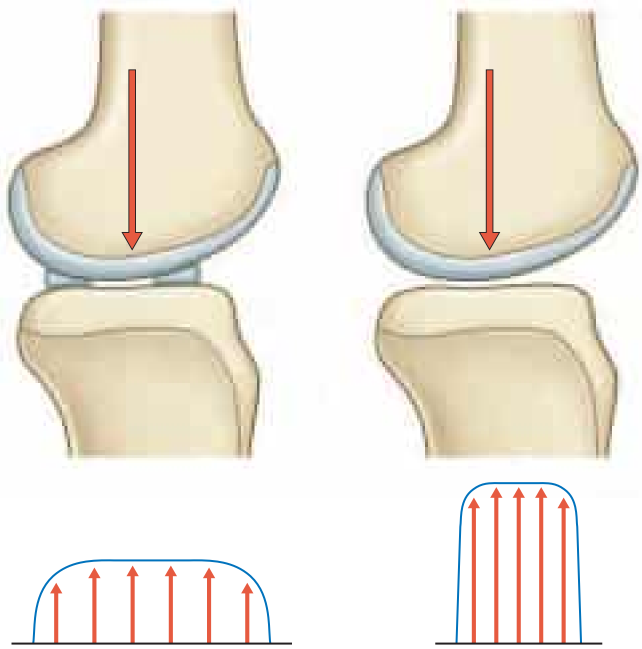


Fig. 82.22 The function of the menisci in increasing articular conformity, with increased peak pressure before and after meniscectomy.

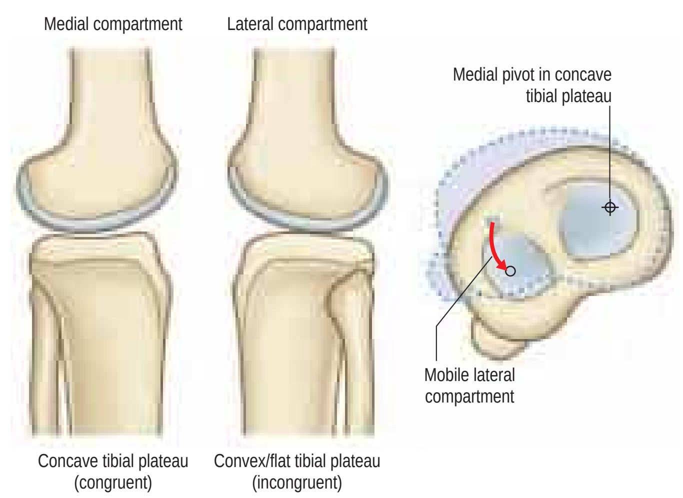


Fig. 82.21 The shape of the articular surfaces providing mobility of the lateral compartment.

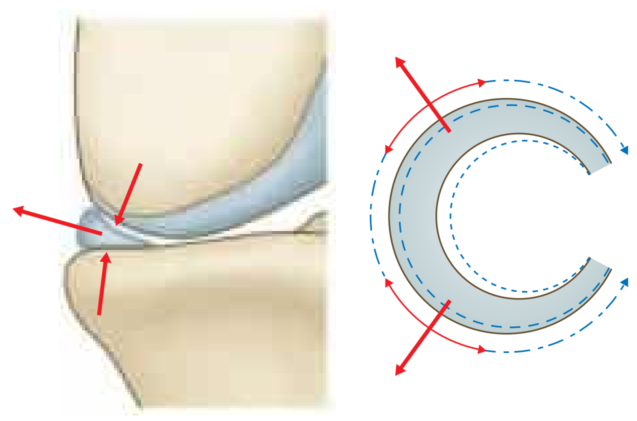


Fig. 82.23 The load-carrying mode of the menisci. Conversion of axial load into meniscal hoop stresses.

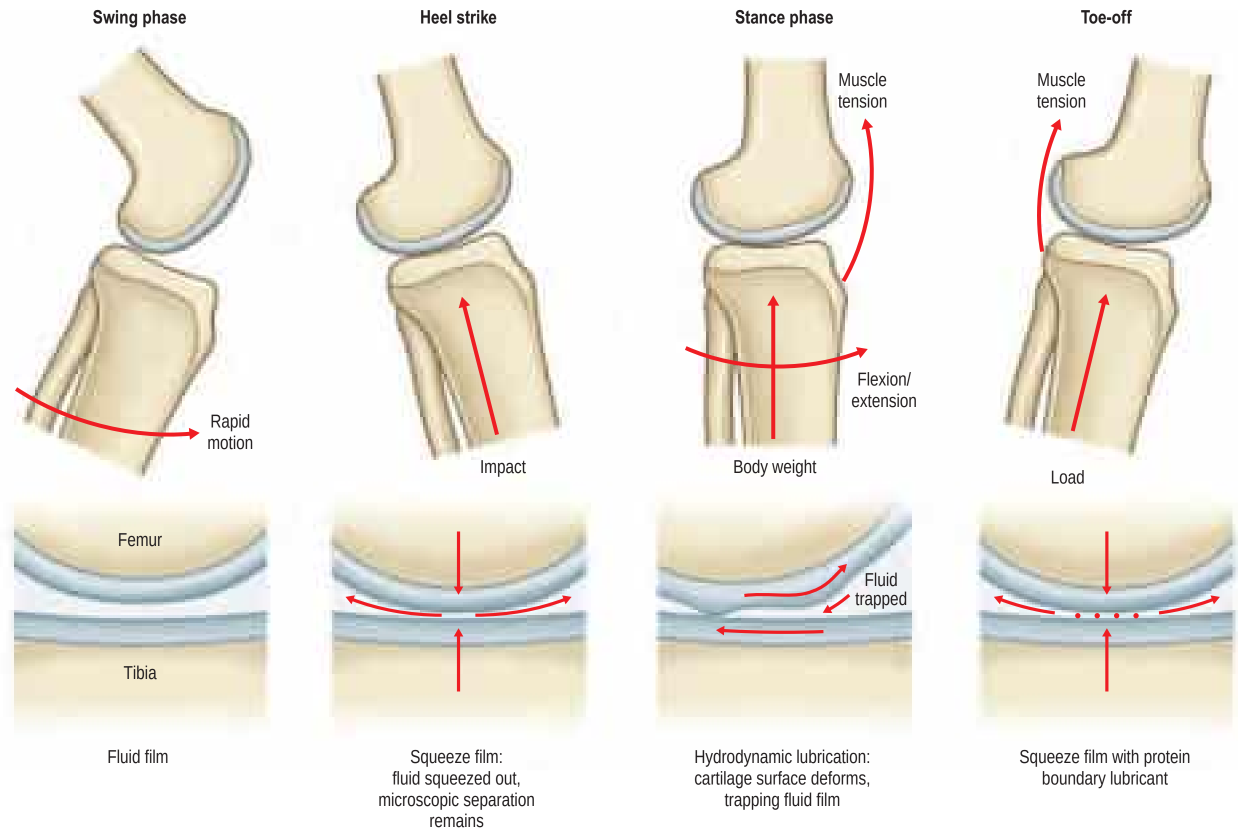


Fig. 82.24 The different modes of knee joint motion and lubrication when walking.

loaded lightly, and have high relative velocity while the knee flexes and extends. This allows the synovial fluid to separate the surfaces, giving fluid film lubrication, with very low friction and no wear (Fig. 82.24). At the time of heel strike, a large impact load acts to compress the surfaces together. At a microscopic level, the surfaces do not come into contact because of the squeeze-film effect: in essence, the impact occurs so rapidly (less than 0.1 sec) that the synovial fluid cannot all be squeezed out of the joint space because of its viscosity and the narrowness of the space. In the mid-stance phase, the flexion–extension motion again entrains the synovial fluid in between the joint surfaces, producing what is known as a hydrodynamic effect: the fluid is trapped between the surfaces by the motion and therefore it acts to separate them. Finally, when the foot is pushing off the body weight, there is little motion and the thin fluid film diminishes under the compressive load. If the squeeze-film effect were insufficient, then the joint surfaces would come into direct contact were it not for the fact that the synovial fluid contains large protein molecules that are trapped on the cartilage surfaces when the fluid is expelled. This molecular layer acts as a boundary lubricant, protecting the cartilage in the same way that grease protects a synthetic bearing.

Many aspects of the arthritic breakdown of the cartilage can be explained on the basis of lubrication biomechanics. Thus, in joints affected by osteoarthrosis, synovial fluid is known to have a lower viscosity than that in normal joints. The viscosity is lower still in joints afflicted with rheumatoid arthritis and is unable to prevent attritional damage to the cartilage surfaces.

SOFT TISSUE MECHANICS

The knee joint relies on active (musculotendinous) and passive (ligamentous) restraints to maintain its stability. The muscles provide the loading to move the joint: quadriceps femoris, hamstrings and gastrocnemius control both flexion/extension and medial–lateral rotation. However, they also cause anterior–posterior shear forces that are resisted primarily by the cruciate ligaments. This tethering effect is critical in allowing the joint to move physiologically, maintaining congruency and stability.

The passive stabilizers of the joint act by resisting unwanted displacements between the bones. This may be to control the path of motion or to limit the range of motion. When the muscles or some other external force (due to body weight or impact) cause the bones to displace, the ligaments are stretched, and so develop tensile forces that resist the displacement and allow the joint to maintain its stability. Disruption of any of these passive restraints may cause a mechanical instability, which is an abnormally increased displacement due to an applied force (in biomechanical terms this is called excess laxity).

Primary and secondary restraints

On testing the laxity in any of the degrees of freedom of the joint (e.g. in the anterior drawer test, a bedside clinical test in which the subject is placed in the supine position with the knee to be tested flexed to 80–90° before passive forward traction is applied to the tibia), there are usually combinations of ligaments that are tensed. Some of these are better aligned to resist the applied load or displacement. These are termed primary restraints, and are exemplified by the cruciate ligaments and the tibial and fibular collateral ligaments. Secondary restraints are less well aligned but still have a significant restraining effect. These are exemplified by the menisci and by the meniscofemoral ligaments. In the example in Figure 82.25, the anterior cruciate ligament is well aligned to resist the applied anterior force. With an absent anterior cruciate ligament, the tibial collateral ligament can resist the applied force; however, it does so by being loaded to a much higher level than the original loading on the anterior cruciate ligament. The size of the lines in the vector diagram demonstrate this principle: although joint laxity may remain normal initially following rupture of a primary restraint, it may subsequently result in the overload of a secondary restraint and, ultimately, in further soft tissue failure.

The patellofemoral joint is most heavily loaded during weight-bearing activities when the knee is flexed. Analyses of rising from a chair have predicted that the patellar ligament tension at 90° of knee flexion may be greater than the tibiofemoral joint, which is loaded at the same time (Amis and Farahmand 1996).

In the frontal plane, quadriceps femoris and the patellar ligament tensions combine to cause a lateralizing force vector termed the Q-angle effect. The Q angle is defined as the difference between the resultant force vector of quadriceps femoris, which is normally parallel to the femoral shaft, and the patellar ligament (Fig. 82.26). Clinically, the Q angle is changed by the position of hip rotation, tibial rotation and quadriceps femoris tension. The clinical Q angle is 12–15° (males) and 15–18° (females), which means that there is a greater lateralizing force vector on the patellofemoral joint in females. Contraction of quadriceps femoris, therefore, tends to displace the patella laterally, which is resisted by the geometry of the joint and by the ligaments. Vastus medialis obliquus acts medially and posteriorly as much as it acts proximally, and so its tension helps to resist the Q-angle effect.

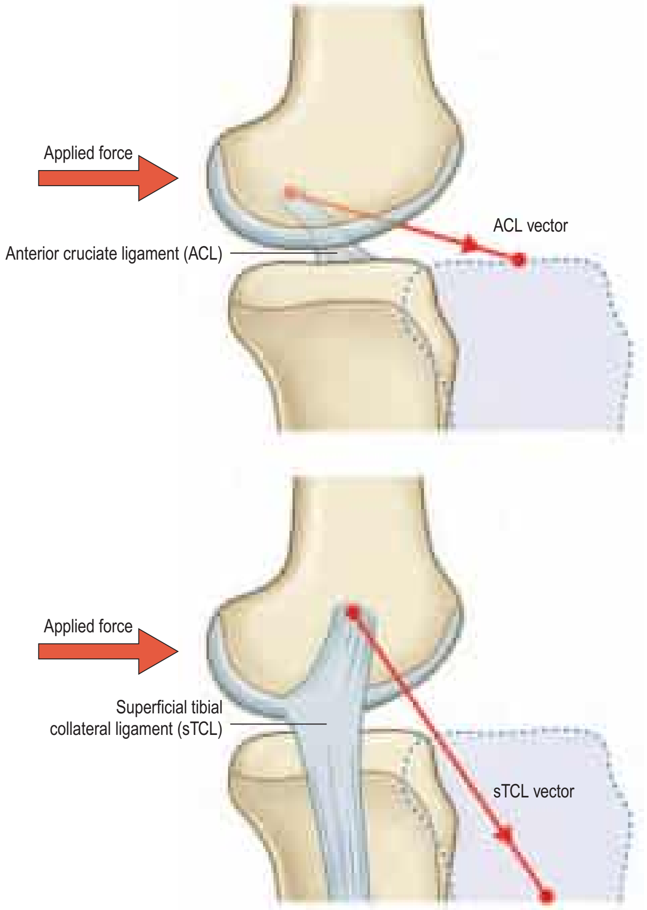
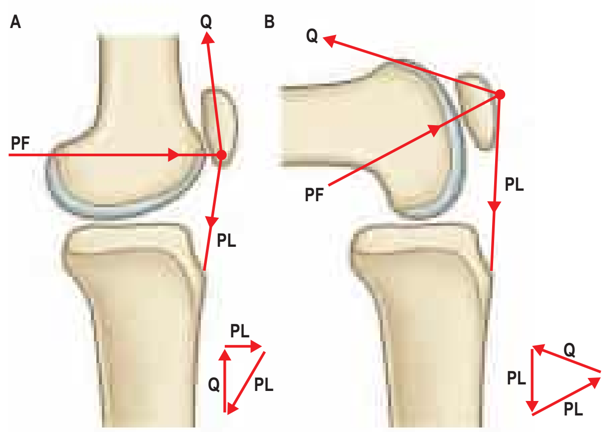


Fig. 82.25 Primary and secondary restraints to anteroposterior forces. The example shows the anterior cruciate ligament (ACL) and superficial part of the tibial collateral ligament (sTCL).



PL = Patellar ligament force **PF** = Patellofemoral joint reaction force **Q** = Quadriceps femoris traction force

Fig. 82.26 The patellofemoral joint force with knee extension (**A**) and with 90° of knee flexion (**B**).

MUSCLES

The majority of muscles that act on the knee joint are described in either Chapter 80 (quadriceps femoris, semimembranosus, biceps femoris, semitendinosus, articularis genus) or on page 1409 (gastrocnemius). Popliteus is described below.

Popliteus

Attachments Popliteus is a flat muscle that forms the floor of the lower part of the popliteal fossa (see Figs 82.12A, 83.4B, 83.9A). It arises within the capsule of the knee joint by a strong tendon, 2.5 cm long, which is attached to a depression at the anterior end of the groove (groove for popliteus) on the lateral aspect of the lateral condyle of the femur. Medially, this tendon is joined by collagenous fibres arising from the arcuate popliteal ligament; the fibrous capsule adjacent to the lateral meniscus; and the outer margin of the meniscus.

Popliteus is attached to the medial aspect of the head of the fibula by the popliteofibular ligament, which passes laterally and inferiorly from the popliteus tendon in a sheet of tissue that is normally about 2 cm^2. This ligament is the single most important stabilizer of the posterolateral region of the knee and resists lateral rotation of the tibia on the femur. Failure to recognize and reconstruct damage to this ligament and to the related ligamentous structures is the most common reason for a poor result from an otherwise well-performed operation for repair of ruptured cruciate ligaments.

Fleshy fibres expand from the inferior limit of the tendon to form a somewhat triangular muscle that descends medially to be inserted into the medial two-thirds of the triangular area above the soleal line on the posterior surface of the tibia, and into the tendinous expansion that covers its surface.

An additional head may arise from the sesamoid bone in the lateral head of gastrocnemius. Very rarely, two other muscles may be found deeply situated behind the knee. Popliteus minor runs from the posterior surface of the lateral tibial condyle, medial to plantaris, to the oblique popliteal ligament. Peroneotibialis runs deep to popliteus from the medial side of the fibular head to the upper end of the soleal line.

Relations Popliteus is covered posteriorly by a dense aponeurotic expansion, which is largely derived from the tendon of semimembranosus. Gastrocnemius, plantaris, the popliteal vessels and the tibial nerve all lie posterior to the expansion. The popliteal tendon is intracapsular and is deep to the fibular collateral ligament and the tendon of biceps femoris. It is invested on its deep surface by synovial membrane, and grooves the posterior border of the lateral meniscus and the adjoining part of the tibia before it emerges inferior to the posterior band of the arcuate ligament. This region is called the 'popliteus hiatus'; the lateral meniscus has no peripheral bony attachment here and is therefore rather mobile.

Vascular supply The arterial supply of popliteus is derived mainly from the inferior medial and lateral genicular arteries. The latter may cross either superficial or deep to the muscle. There are additional contributions from the nutrient artery of the tibia (from the posterior tibial artery), the proximal part of the posterior tibial artery, and the posterior tibial recurrent artery.

Innervation Popliteus is innervated by a branch of the tibial nerve (L4, 5 and S1), which winds around the distal border of popliteus and enters the anterior surface of the muscle; this nerve also innervates the superior tibiofibular joint and the interosseous membrane of the leg.

Actions Popliteus rotates the tibia medially on the femur or, when the tibia is fixed, rotates the femur laterally on the tibia. It is usually regarded as the muscle that 'unlocks' the joint at the beginning of flexion of the fully extended knee; electromyography supports this view. Its connection with the arcuate popliteal ligament, fibrous capsule and lateral meniscus has led to the suggestion that popliteus may retract the posterior horn of the lateral meniscus during lateral rotation of the femur and flexion of the knee joint, thus protecting the meniscus from being crushed between the femur and the tibia during these movements. The muscle is markedly active in crouching, perhaps to provide stability as the tibia rotates medially during flexion of the knee. However, the main function is likely to be provision of dynamic stability to the posterolateral part of the knee by preventing excessive lateral rotation of the tibia, partly by its direct action, but more significantly by tensing the popliteofibular ligament.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

There is an intricate arterial anastomosis around the patella and the femoral and tibial condyles (see Fig. 78.4). A superficial arterial network spreads between the fascia and skin around the patella and in the fat deep to the patellar ligament. A deep arterial network lies on the femur and tibia near the adjoining articular surfaces, and supplies the bone, articular capsule, synovial membrane and cruciate ligaments (see Fig. 82.1). The vessels involved are the superior, middle and inferior genicular branches of the popliteal artery; descending genicular branches of the femoral artery and descending branch of the lateral circumflex femoral artery; circumflex fibular artery; and anterior and posterior tibial recurrent arteries. For details, consult Scapinelli (1968).

Variations in the arterial anatomy of the lower limb

Most anatomical variations in the arterial pattern of the lower limb arteries cause no symptoms. They are usually incidental findings in the dissection room, or may come to light in the course of angiographic examination. Some variations in vascular anatomy may be symptomatic and may necessitate surgical correction; other variations, while asymptomatic, may influence technical considerations during vascular surgical procedures.

Anatomical variations of the femoral, profunda femoris, anterior and posterior tibial and fibular arteries have been described elsewhere in this Section.

Popliteal artery

The popliteal artery is the continuation of the femoral artery and crosses the popliteal fossa (see Figs 82.2, 82.4). It descends laterally from the opening in adductor magnus to the femoral intercondylar fossa, inclining obliquely to the distal border of popliteus, where it divides into the anterior and posterior tibial arteries. This division usually occurs at the proximal end of the interosseous space between the wide tibial metaphysis and the slender fibular metaphysis. The artery is relatively tethered at the adductor hiatus and again distally by the fascia related to soleus, and is therefore susceptible to damage following knee injuries, e.g. dislocation. Aneurysms of the popliteal artery are not uncommon.

Variations in the popliteal artery

The popliteal artery may occasionally bifurcate more superiorly and divide into its terminal branches proximal to popliteus, or it may trifurcate into anterior and posterior tibial and fibular arteries. Either the anterior tibial or the posterior tibial artery may be reduced or increased in size. The size of the fibular artery is usually inversely related to the size of the anterior and posterior tibial arteries. Rarely, the anterior tibial artery is the source of the fibular artery, a variation that is almost always associated with a high bifurcation of the popliteal artery.

The popliteal and inferior gluteal arteries may be joined by a large anastomotic vessel; in such cases, the femoral artery is hypoplastic (Kawashima and Sasaki 2010). When this occurs, the popliteal artery usually has an abnormal relationship to popliteus, running deep to the muscle before dividing into its terminal branches. The popliteal artery may pass medially beneath the medial head of gastrocnemius, or may pass beneath an aberrant band of muscle in the popliteal fossa; in either case, contraction of the muscles may occlude the artery – a condition that may present with claudication on exertion. The popliteal vein is usually superficial and adjacent to the artery, but it may run deep to the artery, or be separated from it by a slip of muscle derived from the medial head of gastrocnemius.

Relations

Anteriorly, from proximal to distal, are fat covering the femoral popliteal surface, the capsule of the knee joint, and the fascia of popliteus. Posteriorly are semimembranosus (proximally) and gastrocnemius and plantaris (distally). In between, the artery is separated from the skin and fasciae by fat and crossed from its lateral to its medial side by the tibial nerve and popliteal vein; the vein lies between the nerve and artery and is adherent to the latter. Laterally are biceps femoris, the tibial nerve, popliteal vein and lateral femoral condyle (all proximal), and plantaris and the lateral head of gastrocnemius (distal). Medially are semimembranosus and the medial femoral condyle (proximal), and the tibial nerve, popliteal vein and medial head of gastrocnemius (distally). The relations of the popliteal nodes are described below.

Branches (other than terminal)

Genicular anastomosis There is an intricate arterial anastomosis around the patella and femoral and tibial condyles (see Fig. 82.1). A superficial network spreads between the fascia and skin around the patella and in the fat deep to the patellar ligament. A deep network lies on the femur and tibia near their adjoining articular surfaces, and supplies the bone, articular capsule and synovial membrane. The vessels involved in this anastomosis are superior medial and lateral genicular; inferior medial and lateral genicular; descending genicular; descending branch of the lateral circumflex femoral; circumflex fibular; and anterior and posterior tibial recurrent arteries.

Superior genicular arteries The superior genicular arteries branch from the popliteal artery, curving round proximal to both femoral condyles to reach the anterior aspect of the knee. The superior medial genicular artery lies under semimembranosus and semitendinosus, proximal to the medial head of gastrocnemius and deep to the tendon of adductor magnus. It divides into a branch to vastus medialis that anastomoses with the descending genicular and inferior medial genicular arteries, and a branch that ramifies on the femur and anastomoses with the superior lateral genicular artery. Its size varies inversely with that of the descending genicular artery. The superior lateral genicular artery passes under the tendon of biceps femoris, pierces the lateral intermuscular septum and divides into superficial and deep branches. The superficial branch supplies vastus lateralis and anastomoses with the descending branch of the lateral circumflex femoral and inferior lateral genicular arteries, while the deep branch anastomoses with the superior medial genicular artery, forming an anterior arch across the femur with the descending genicular artery. The superficial branch is vulnerable if the lateral patellar retinaculum is divided surgically.

Middle genicular artery The middle genicular artery is small. It arises from the popliteal artery near the midpoint of the posterior aspect of the knee joint. It pierces the oblique popliteal ligament to supply the cruciate ligaments and synovial membrane.

Inferior genicular arteries The inferior medial and lateral genicular arteries arise from the popliteal artery deep to gastrocnemius. The inferior medial genicular artery is deep to the medial head of gastrocnemius. It descends along the proximal margin of popliteus, which it supplies; passes inferior to the medial tibial condyle, under the tibial collateral ligament; and then ascends anteromedial to the knee joint at the anterior border of the tibial collateral ligament. It supplies the joint and the tibia, and anastomoses with the inferior lateral and superior medial genicular arteries and with the anterior tibial recurrent artery and saphenous branch of the descending genicular artery. The inferior lateral genicular artery runs laterally across popliteus and forwards over the head of the fibula to the front of the knee joint, passing under the lateral head of gastrocnemius, the fibular collateral ligament and the tendon of biceps femoris. Its branches anastomose with the inferior medial and superior lateral genicular arteries; anterior and posterior tibial recurrent arteries; and the circumflex fibular branch of the posterior tibial artery.

Cutaneous branches: the superficial sural arteries The superficial sural arteries are three vessels that leave the popliteal artery, or its side branches, descend between the heads of gastrocnemius and perforate the deep fascia to supply the skin on the back of the leg. The central or median vessel is usually larger than the medial or lateral vessels, and usually accompanies the sural nerve.

Fasciocutaneous free and pedicled flaps may be raised on the superficial sural arteries.

Superior muscular branches The superior muscular branches are two or three vessels that arise proximally and pass to adductor magnus and the posterior thigh muscles. They anastomose with the termination of the profunda femoris artery.

Sural arteries The two sural arteries are large and arise behind the knee joint to supply gastrocnemius, soleus and plantaris. They are used in gastrocnemius musculocutaneous flaps.

VEINS

The veins draining the knee correspond in name to the arteries and run with them; the named smaller veins drain into the popliteal and femoral veins.

Popliteal vein

The popliteal vein ascends through the popliteal fossa to the opening in adductor magnus, where it becomes the femoral vein (see Figs 78.8, 78.9B, 80.30). Its relationship to the popliteal artery changes as the vein ascends: distally, it is medial to the artery; between the heads of gastrocnemius, it is superficial (posterior) to the artery; and proximal to the knee joint, it is posterolateral to the artery. Its tributaries are the short saphenous vein; veins corresponding to branches of the popliteal artery; and muscular veins, including a large branch from each head of gastrocnemius. There are usually four or five valves in the popliteal vein.

Long saphenous vein

The course of the long saphenous vein is described on pages 1370–1371.

Short saphenous vein

The short saphenous vein (small saphenous vein) begins posterior to the lateral malleolus as a continuation of the lateral marginal vein (see Fig. 78.9B). In the lower third of the calf, it ascends lateral to the calcaneal tendon, lying on the deep fascia and covered only by subcutaneous tissue and skin. Inclining medially to reach the midline of the calf, it penetrates the deep fascia, within which it ascends on gastrocnemius, only emerging between the deep fascia and gastrocnemius gradually at about the junction of the middle and proximal thirds of the calf (usually well below the lower limit of the popliteal fossa). Continuing its ascent, it passes between the heads of gastrocnemius and proceeds to its termination in the popliteal vein, 3–7.5 cm above the knee joint.

Tributaries

The short saphenous vein connects with deep veins on the dorsum of the foot, receives many cutaneous tributaries in the leg, and sends several communicating branches proximally and medially to join the long saphenous vein. Sometimes a communicating branch ascends medially to the accessory saphenous vein: this may be the main continuation of the short saphenous vein. In the leg, the short saphenous vein lies near the sural nerve and contains 7–13 valves, with one near its termination. Its mode of termination is variable: it may join the long saphenous vein in the proximal thigh or it may bifurcate, one branch joining the long saphenous vein and the other joining the popliteal or deep posterior femoral veins. Sometimes it drains distal to the knee in the long saphenous or sural veins.

LYMPHATIC DRAINAGE

Lymphatic drainage is to the popliteal nodes (see Fig. 78.10). Most of the lymph vessels accompany the genicular arteries; some vessels from the joint drain directly into a node between the popliteal artery and the posterior capsule of the knee joint.

Popliteal nodes

There are usually six small lymph nodes embedded in the fat of the popliteal fossa. One, near the termination of the short saphenous vein, drains the superficial region served by this vessel. Another lies between the popliteal artery and the posterior aspect of the knee, receiving direct vessels from the knee joint and those accompanying the genicular arteries. The remainder flank the popliteal vessels, receiving trunks that accompany the anterior and posterior tibial vessels. Popliteal lymphatic vessels ascend close to the femoral vessels to reach the deep inguinal nodes; some may accompany the long saphenous vein to the superficial inguinal nodes.

INNERVATION

The knee joint is innervated by branches from the obturator, femoral, tibial and common fibular nerves (see Fig. 78.11) (Freeman and Wyke 1967). The articular branch of the obturator nerve is the terminal branch of its posterior division. Muscular branches of the femoral nerve, especially to vastus medialis, supply articular branches to the knee joint. Genicular branches from the tibial and common fibular nerves accompany the genicular arteries, and those from the tibial nerve travel with the medial and middle genicular arteries, while those from the common fibular nerve travel with the lateral genicular and anterior tibial recurrent arteries.

The femoral and obturator nerves are described on page 1372, and the tibial and common fibular nerves are described on page 1415.

Saphenous nerve

The saphenous nerve is the largest and longest cutaneous branch of the femoral nerve and the longest nerve in the body (see Figs 78.11, 80.32). It descends lateral to the femoral artery in the femoral triangle and enters the adductor canal, where it crosses anterior to the artery to lie medial to it. At the distal end of the canal, it leaves the artery and emerges through the aponeurotic covering with the saphenous branch of the descending genicular artery. As it leaves the adductor canal, it gives off an infrapatellar branch that contributes to the peripatellar plexus and then pierces the fascia lata between the tendons of sartorius and gracilis, becoming subcutaneous to supply the skin anterior to the patella. It descends along the medial border of the tibia with the long saphenous vein and divides distally into a branch that continues along the tibia to the ankle and a branch that passes anterior to the ankle to supply the skin on the medial side of the foot, often as far as the first metatarsophalangeal joint. The saphenous nerve connects with the medial branch of the superficial fibular nerve. Near the mid-thigh, it provides a branch to the subsartorial plexus. The nerve may become entrapped as it leaves the adductor canal.

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